LC Temperature and Matter Lecture Notes

Year 1 Semester 2

Ash Stewart

MSci Physics with Particle Physics and Cosmology

School of Physics and Astronomy
University of Birmingham

Table of Lectures

Tue 20 Jan 2026 12:00

Lecture 1 - Matter I: Welcome and Intro

1 Course Welcome

- Study guide and full revision notes (from Prof. Mike Gunn) on Canvas.
  - The course has changed a bit recently, so the full revision notes have additional theory/content which isn't examinable.
- Series of non-assessed problems also on Canvas.

1.1 Course Aims

We want to be able to explain the properties of matter (in the macroscopic world) of various kinds, based on their microscopic behaviour. We also want to be able to understand more about the microscopic behaviour of matter from macroscopic observables.

1.2 Recommended Reading

- Young and Freedman, Ch14 and Ch17-19.
- Properties of Matter, Flowers and Mendoza.
- Introduction to Thermal Physics, Adkins.

2 Intro to Matter

2.1 States of Matter

We want to move away from a simple ”cold stuff is a solid, warmer stuff is a liquid and hotter stuff is a gas” by also considering how different materials change state as a function of both pressure and temperature:

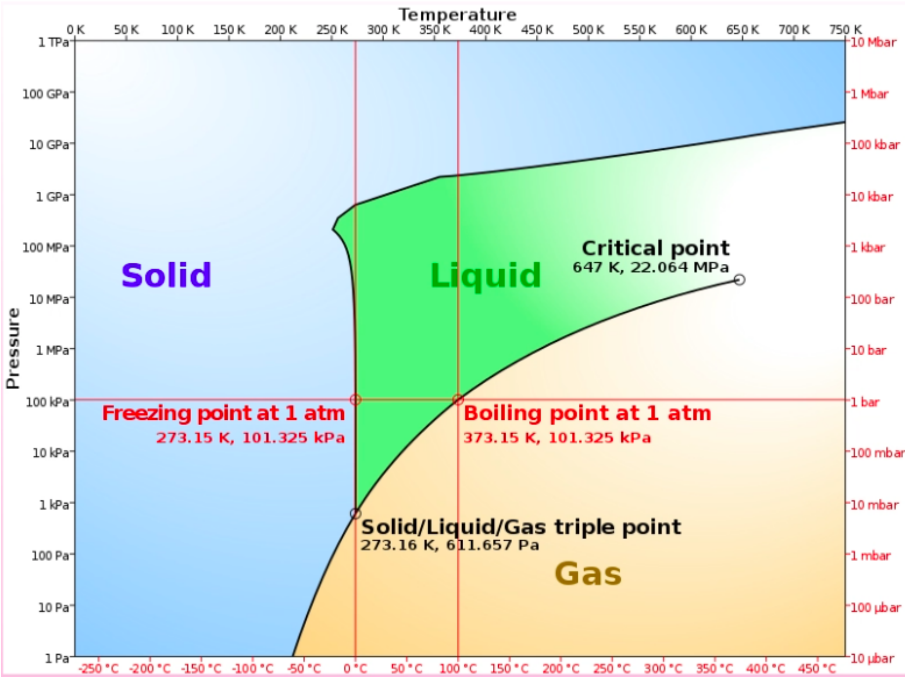

Figure 1.1: Phase Diagram for Water

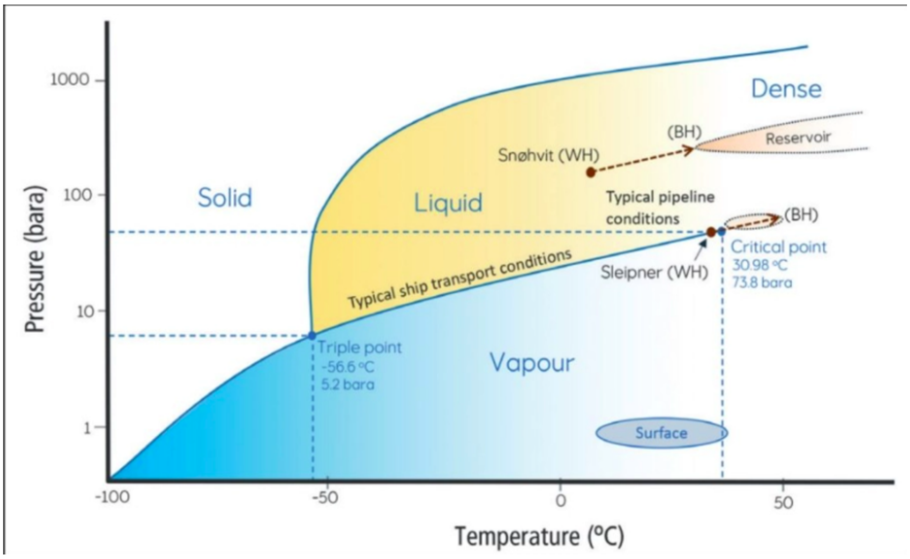

Figure 1.2: Phase Diagram for CO2

At “Standard Pressure and Temperature” (at 20°C and 1ATM), we see that increasing pressure of C02 takes it to a liquid state. However, at lower temperatures, decreasing the pressure can cause it to sublimate straight from a solid to a gas, without a liquid intermediary.

Taking water as an example, pushing the pressure/temperature causes another high-energy state altogether called a “supercritical fluid”. The gradient for the initial vertical portion of the solid/liquid boundary around the freezing point is slightly negative, so applying pressure to ice on this boundary causes it to become liquid. This is uncommon for many materials.

This is because water is weird, where the solid form is less dense than the liquid form. This is how ice skating works, where applying pressure (from a skate) causes a thin upper layer of ice to melt. This thin water layer is what the ice skate can travel on.

Every material also has a “triple point” where the material can act simultaneously in all three standard states of matter.

Thu 22 Jan 2026 13:00

Lecture 2 - Matter II: Atomic Physics and LJP

1 Atomic Sizes

As we increase proton number, we might expect that the atomic radius increases (as there's physically more stuff). We might expect that adding more electrons would expect a corresponding increase in the atomic radius to fit these extra electrons.

This doesn't actually prove to be true, and atomic size and properties are generally determined by the outermost electrons, and atomic radius stays mostly constant.

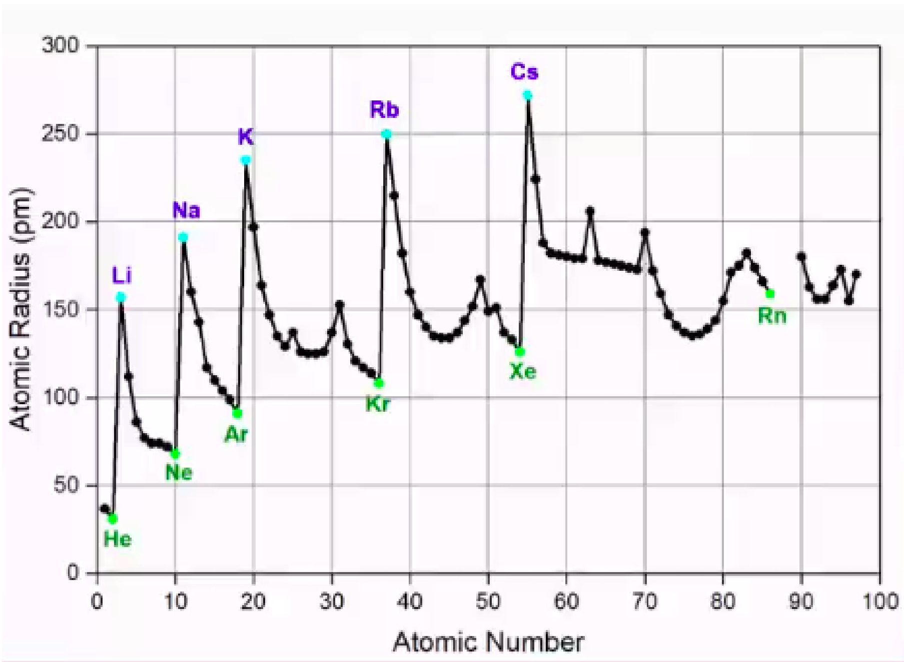

Figure 2.1: Variation of Atomic Radius with Atomic Number

We do see a large increase in atomic radius for the group one metals, where the outermost electron is very weakly attracted to the nucleus (is easily ionised). For an exam, we can consider atomic radius to be approximately 10^{-10}.

We see a similar trend when considering ionisation energy (the energy required to remove the outermost electron):

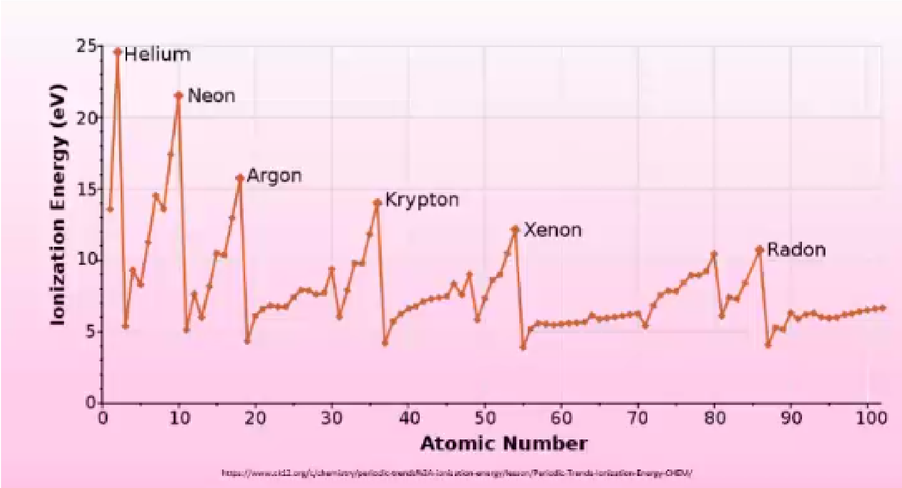

Figure 2.2

This is approximately the same for the all atomic numbers, with the exception of noble gases (which are extremely difficult to ionise as a result of having full capacity outer shells).

This doesn't apply however to the nuclear radius, R, as the nucleus is compressed to a much higher degree, so there's not space to fit additional nucleons without increasing radius. For atomic radius A, and modelling the nucleus as a sphere, we get:

$$R = r_0 A^{\frac{1}{3}}$$

Where $r_0 = 1.2 \times 10^{-15} m = 1.2\text{fm}$ is a constant.

2 Atomic Masses

Given the extremely small mass scales, we generally give masses of atoms or molecules in terms of atomic mass units, amu, where:

$$1\text{amu} = 1.66 \times 10^{-27}\text{kg}$$

This is defined as 1/12th the mass of a Carbon-12, ^{12}C atom. The atomic mass of Carbon-12 is thus 12amu, however a proton and neutron each have mass 1.007amu and 1.009amu respectively. Why is there a discrepancy if we add the masses of the individual nucleons?

This is because, when combining these nucleons into a nucleus, some of this mass is converted into energy. This is called "binding energy" and is used to hold the nucleus together.

2.1 Moles

A mole (mol) is an SI base unit used to described the amount of a substance. 1 mole of a substance is defined by Avagadro's number, $N_A = 6.022 \times 10^{23}$, where one mole has N_A atoms/molecules. We again use Carbon 12, as 12 grams of Carbon-12 has N_A atoms.

We further have:

$$\text{amount, mol} = \frac{N}{N_A}$$

$$\text{amount, mol} = \frac{\text{Mass}}{\text{Atomic Mass}}$$

For gasses, one mole occupies exactly 22.4L or 22.4dm^3 at standard temperature and pressure (0°C, 10^5Pa = 1atm), regardless of the gas in consideration.

3 Rayleigh's Oil Drop Measurement

Rayleigh conducted an experiment where he took a large volume of water, and placed a drop of olive oil (of known volume V) on top. He postulated that, due to the nature of olive oil being long-chained hydrocarbons, the oil would spread out perfectly thinly, and produce a layer exactly one atom tall.

He said that this would form a circular layer of radius R and height d. If we take the film to be a cylinder (and d is too small to measure directly), we can measure R and use the known volume to determine d:

$$V = \pi R^2 d \implies d = \frac{V}{\pi R^2}$$

When he carried this out, he determined the atomic radius [1], as $\sim$ 1nm, i.e. being one order of magnitude out. This wasn't bad for how rudimentary the experiment was!

4 Latent Heat

We know that there has to be some kind of force between two atoms. If we consider a diatomic molecule, separating these atoms from each other to infinity requires some amount of energy ϵ. Because there are two atoms, the work done per atom is $\epsilon/2$.

Lets consider a more complex example with many more atoms (i.e. converting a solid to liquid):

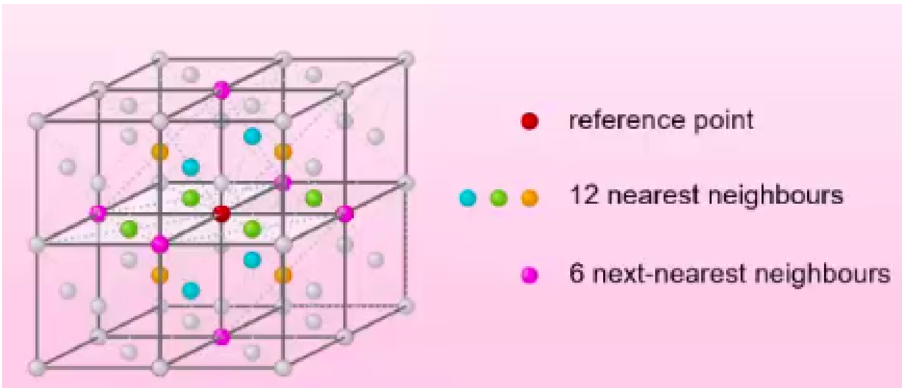

Figure 2.3

In this case, extracting the red atom from all its 12 nearest neighbours will take $12\epsilon/2$. We should additionally consider the 6 next nearest neighbours and so and so on. Each increase in number of nth next nearest neighbours has a decreasing impact in the force required, so leads to diminishing returns. In this course, we often ignore he next nearest neighbours and further. If asked to estimate, we use $6-10$ for a liquid, and $8-12$ for a solid.

While we're considering this in a solid, lets use it as an approximation for a liquid. The latent heat of vaporisation L(Jkg) is the amount of heat required to completely vaporise a liquid into a gas.

$$L = 12\epsilon/2 \times \frac{\text{number of atoms}}{\text{mass of material, kg}}$$

And the number/mass is given by:

$$\frac{\text{kg}}{\text{atoms}} = \rho \times d^3 \implies \frac{\text{atoms}}{\text{kg}} = \frac{1}{\rho d^3}$$

Hence:

$$d = \sqrt[3]{\frac{1}{\text{atoms per kg} \times \rho}}$$

[1]Strictly, an upper limit for the radius

5 Forces Between Atoms

What causes the forces between atoms? It can't be the electrostatic (coulomb) force, as atoms are electrically neutral.

It can't be the nuclear force, as the range is too small. Likewise, it can't be the weak force as this mediates the wrong particles and has an even smaller range.

Finally, it can't be gravitational as atoms have far too low a mass for the gravitational force to have any real impact.

Even though the electrostatic component has no impact, the electromagnetic force as a whole can.

5.1 Water Example

Oxygen is slightly electronegative, while hydrogen is not. This means that if we have a H_2O water molecule, the electrons will be pulled ever so slightly closer to the oxygen and away from the hydrogen. Across the whole atom, this causes a slight charge on each side of the atom:

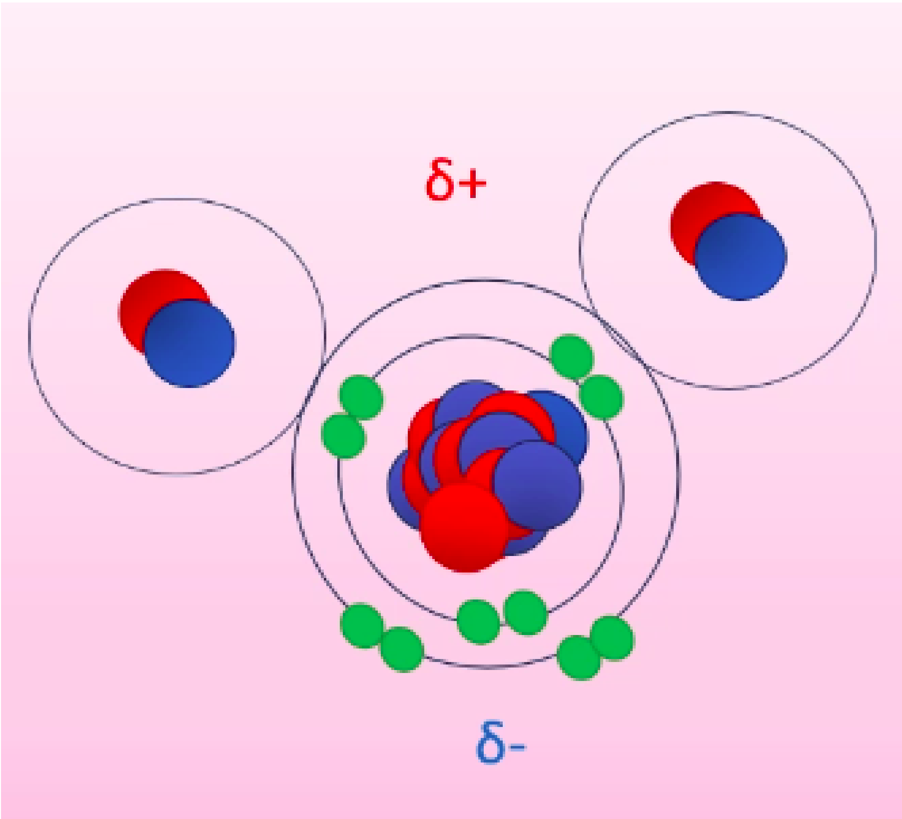

Figure 2.4

Water is "polar molecule" and forms a permanent dipole, where the positive and negative poles of molecules attract.

6 "Normal" Atoms

This only works for some atoms. A block of iron for example cannot be explained by this, as all the atoms are the same, so have equivalent electronegativity.

Here, fluctuations in the electron cloud cause an instantaneous dipole to form. This may be very brief, but lasts for sufficiently long to impact another atom. This atom has electron distribution impacted by this force, causing it to be become a dipole. This creates a field which impacts the original atom etc. This causes the atoms to be stuck as instantaneous dipoles, resulting in a net attractive force called the Van der Waal's force.

As the electric field due to a dipole goes as $|E| \propto 1/r^3$, the induced dipole in the firs atom depends on the strength of its dipole field and the size of the induced dipole in the second atom:

$$|E| = \frac{1}{r^3}\frac{1}{r^3} = \frac{1}{r^6}$$

The force induced by these dipoles explains attraction for non-polar molecules. Force between two atoms is considered positive (by convention) if they repel and negative if they attract.

However, the fact that atoms have a consistent interatomic spacing and do not collapse into black holes suggests that there is some kind of repulsive force that keeps them at fixed distances.

We can infer that this force must be very strong at small distances, but much weaker at larger distances.

7 LJ Potential

We can divide particles into fermions (half integer spin) and bosons (integer spin). The wave fuction of a boson is symmetric about y, so overlapping two bosons causes superposition. Fermions are antisymmetric (i.e. a sine wave). This means that two fermions in the same position causes destructive interference, with a net wave function of 0.

This means that two fermions in the same quantum state (same wave function) cannot share a position, and have a force repelling each other should they try. We approximate this as a potential of the form:

$$V \propto \frac{1}{r^{12}}$$

Lennard-Jones combined these, describing the net potential of the form:

$$V_{LJ} = \frac{A}{r^{12}} - \frac{B}{r^6}$$

Tue 27 Jan 2026 12:00

Lecture 3 - Matter III: Lennard-Jones Potential

1 Potential Distance Plots

If the values of the constants A, B are known, the general form of a Lennard-Jones potential plot is:

$$V_{\text{LJ}} = \underbrace{\frac{A}{r^{12}}}_{\text{+ so repulsive}} - \underbrace{\frac{B}{r^{6}}}_{\text{- so attractive}}$$

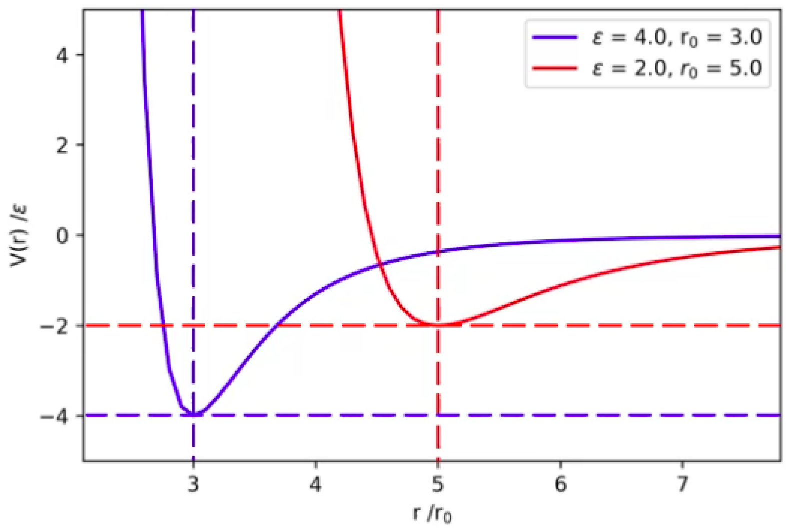

Figure 3.1

The potential is at some baseline minimum value $-\epsilon$ at distance r_0.

For our LJP force, we require:

- Repulsive at very short distances (smaller than interatomic spacing).
- Attractive force for intermediate distances (larger than interatomic spacing).
- No force as $r \to \infty$.
- The atoms equilibrium point is roughly the size of the interatomic spacing in liquids and solids.

In order to describe this more effectively, we need to find expressions for A, B. We do this based on the earlier initial conditions of $V_{LJ} = -\epsilon$ at $r = r_0$

2 Deriving A and B

At the equilibrium point $r = r_0$, we have:

$$V_{LJ} = -\epsilon$$

$$F = 0$$

We can get an expression for force by $F = -dV/dr$:

$$F = -\left(-\frac{12A}{r^{13}} + \frac{60}{r^7}\right) = \frac{12A}{r^13} - \frac{6B}{r^7}$$

At equilibrium:

$$F = 0 = \frac{12A}{r_0^{13}} - \frac{6B}{r_0^7}$$

Disregarding trivial case where $r_0 = 0$, as there cannot be any atoms in this case. Therefore, $r_0 \neq 0$:

$$12A = 6Br_0^6$$

$$\frac{2A}{B} = r_0^6 \implies r_0 = \sqrt[6]{\frac{2A}{B}}$$

And considering potential at equilibrium:

$$-\epsilon = \frac{A}{r_0^{12}} - \frac{B}{r_0^6}$$

$$\implies -\epsilon = \frac{A}{\left(\frac{2A}{B}\right)^2} - \frac{B}{\left(\frac{2A}{B}\right)}$$

$$\implies -\epsilon = \frac{B^2}{4A} - \frac{B^2}{2A}$$

$$\epsilon = \frac{B^2}{2A} - \frac{B^2}{4A} = \frac{B^2}{4A}$$

Putting it all together:

$$\epsilon = \frac{B^2}{4A} \tag{1}$$

$$r_0^6 = \frac{2A}{B} \tag{2}$$

Hence:

$$\frac{2A}{r_0^6} = B \implies \epsilon = \frac{(2A)^2/(r_0^6)^2}{4A}$$

$$A = \epsilon r_0^{12}$$

$$B = 2\epsilon r_0^6$$

So the final functional form of the LJP is:

$$V_{LJ} = \epsilon\left[\left(\frac{r_0}{r}\right)^{12} - 2\left(\frac{r_0}{r}\right)^6\right]$$

Sometimes in textbooks we see:

$$V_{LJ} = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - 2\left(\frac{\sigma}{r}\right)^6\right]$$

Here, σ is the distance where potential is zero, rather than r_0 being the distance where force is zero (potential is $-\epsilon$).

3 Full LJP

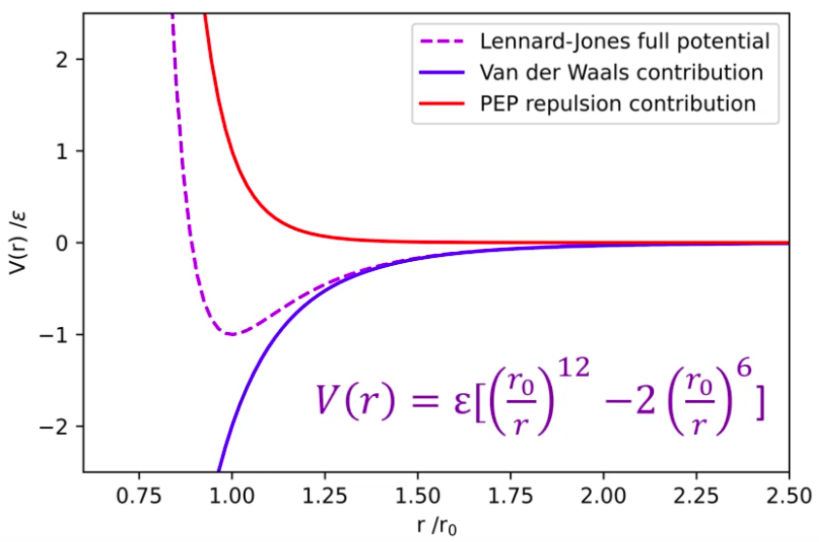

Figure 3.2: The full LJP and the two contributing forces

This gives us the overall LJP that satisfies our three requirements:

- At distances below r_0, the Pauli Exclusion Principle force dominates, causing net repulsion.
- As $r \to \infty$, both components (and the overall potential) tend to zero.
- At distances above, but close to r_0, the Van der Waal's force dominates, causing net attraction.

Where $r/r_0 = 1$, we have a stable equilibrium position, needing $-V(r_0) = -\epsilon = U_0$ to break these apart.

3.1 Implications of LJP

Consider a two dimensional atomic lattice:

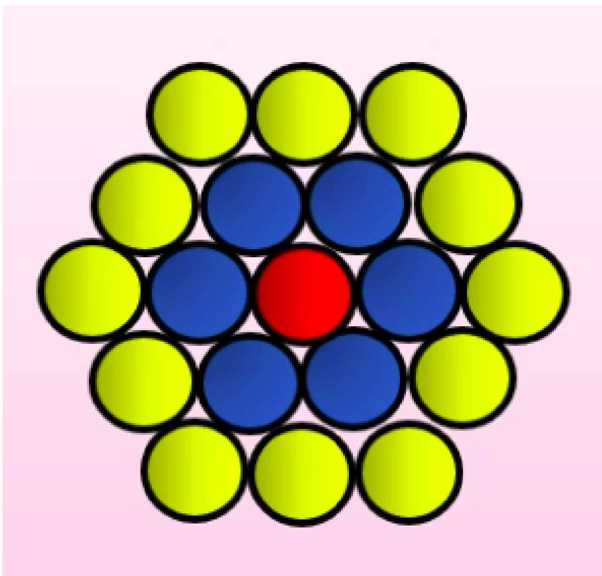

Figure 3.3

The central atom sits in a potential well of approximate depth $N\epsilon$, where N is the number of nearest neighbours. In the equilibrium position, all of these atoms are separated by a distance $r = r_0$.

The next nearest neighbours and beyond do contribute, but to a much smaller degree. It's not zero, but the next nearest neighbours have approximate potential $\epsilon/30$.

We have:

$$V_{LJ} = \epsilon\left[\left(\frac{r_0}{r}\right)^{12} - 2\left(\frac{r_0}{r}\right)^{6}\right]$$

Hence, using $F(r) = -\frac{dV(r)}{dr}$:

$$F(r) = 12\epsilon\left(\frac{r_0^{12}}{r^{13}} - \frac{r_0^6}{r^7}\right)$$

This has a very similar form to the potential:

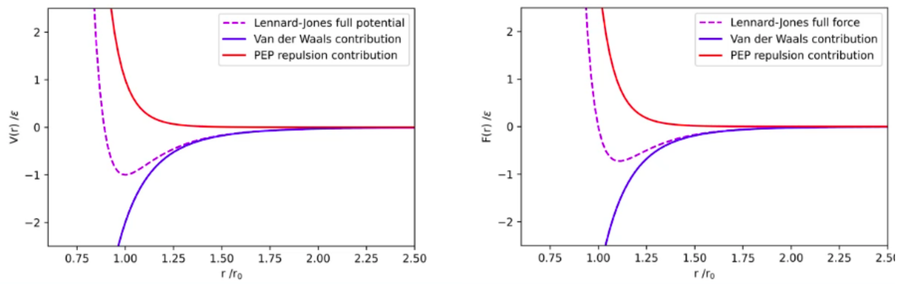

Figure 3.4

Just noting that while potential is $-\epsilon$ at $r_0 = r$, the force is zero.

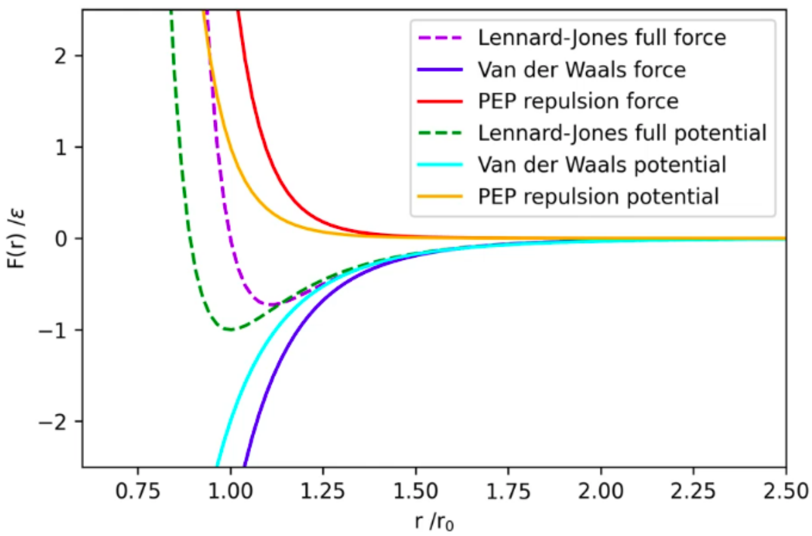

Figure 3.5

3.2 Empirical Data

The LJP provides a surprisingly accurate model for the genuine empirically measured data:

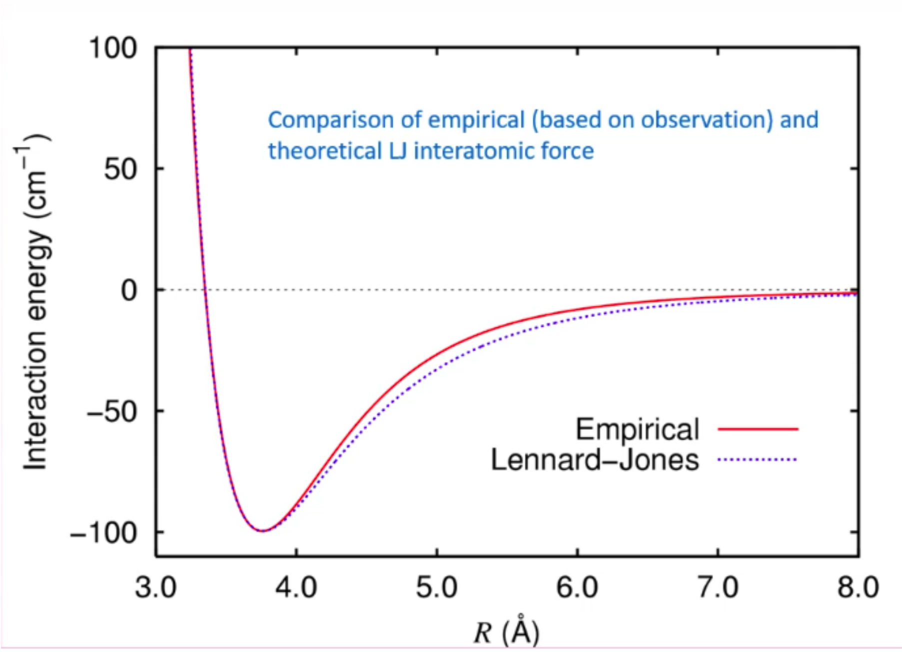

Figure 3.6

It does have limitations though:

- The fact its an infinite range potential means it's very computationally expensive to calculate. We need to truncate the potential once it shrinks below a value below which its negligible.
- For large molecules (such as an amino acid), they are often too large. The distance between the centres of adjacent molecules is too large for the LJP as LJP assumes that they are effectively point like. Amino acids can have forces all the way along their chains and this is not well modelled by the LJP.

Thu 29 Jan 2026 13:00

Lecture 4 - Matter IV: Surface Tension and Capiliary Action

Recap from last time:

- Force and potential are related, with $F = \frac{-dV}{dx}$, with x relating to position.
- At equilibrium, neighbouring atoms sit at distance r_0 (average bond length) from each other. They sit in a potential well of depth $N\epsilon$ where N is the number of nearest neighbours.
- At $x < r_0$, the force is +'ve (repulsive), at $x > r_0$ the force is -'ve (attractive).

1 Coordination Numbers

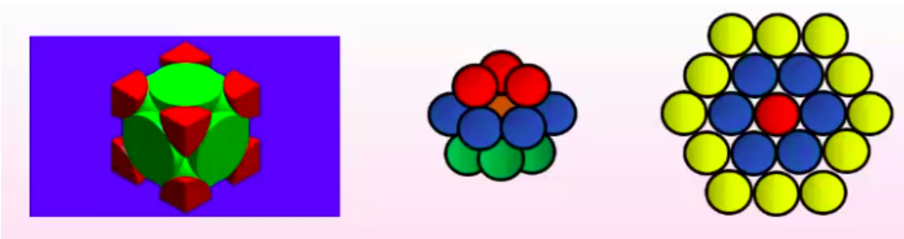

Figure 4.1

In a densely-packed Face-Centred Cubic (left) or Hexagonally Close-Packed Solid (3D, middle - 2D, right) configuration, the maximum coordination number (number of nearest neighbours) is 12. The outermost.

For solids, we expect an interatomic spacing of $\sim 2 \times 10^{-12}$m, for liquids we expect 3×10^{-10}m and for gases at STP we expect about 40×10^{-10}m.

For liquids, we expect a coordination number $n < 10$.

2 Molecular/Structural Binding Energies

To first order (nearest neighbours only), how much energy do we need to remove the innermost atom (orange in the middle figure) from the centre of the structure.

We might expect 12ϵ, the total binding energies of all of the nearest neighbours. Instead however, we are leaving the 12 nearest neighbours in place and only taking one atom to infinity. Since moving both atoms to infinity is given by ϵ, we have:

$$E = \frac{n\epsilon}{2}$$

Where n is the number of nearest neighbours.

To remove one mole of connected atoms, we have:

$$E = \frac{nN_A\epsilon}{2}$$

Again, the factor of two arises to avoid double-counting bonds, i.e. a mole consists of $N_A/2$ pairs of atoms.

3 Latent Heat

Moving macroscopically, we have seen:

$$L = 10\epsilon/2 \times \frac{\text{number of atoms}}{\text{mass of material, kg}}$$

For a liquid of coordination number 10. This is very similar to the just seen:

$$E = \frac{nN_A\epsilon}{2}$$

As N_A gives us atoms per mol, this is effectively identical - one just considers moles and one considers mass in kg.

The molar latent heat equation assume however that the molecular energy comes exclusively from potential energy. This is an approximation that works when kinetic energy is much less than potential energy, i.e. at low temperatures. We can empirically see that molar latent heat varies with temperature:

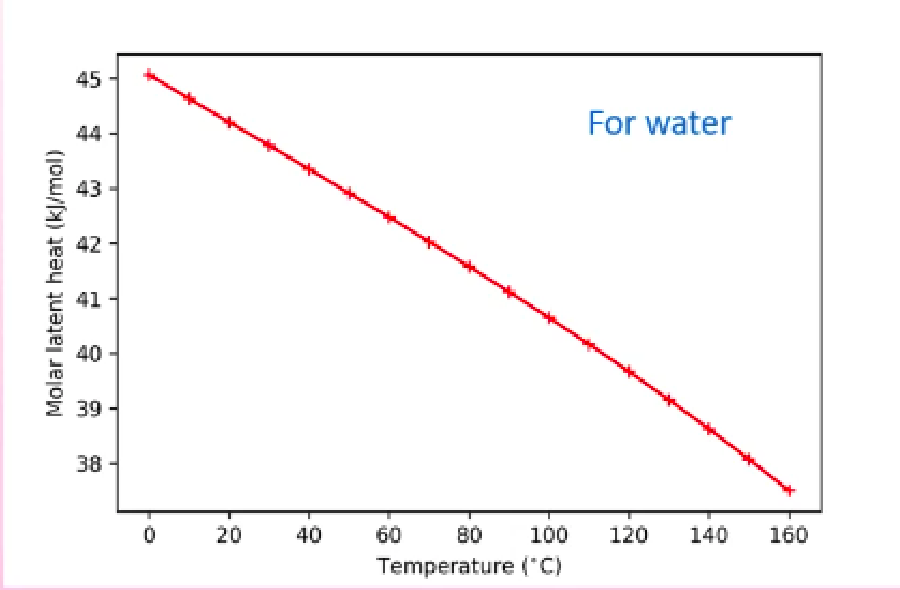

Figure 4.2

This is because, at higher temperatures, the average bond length increases (as the particles have a greater kinetic energy, so vibrate with a larger magnitude):

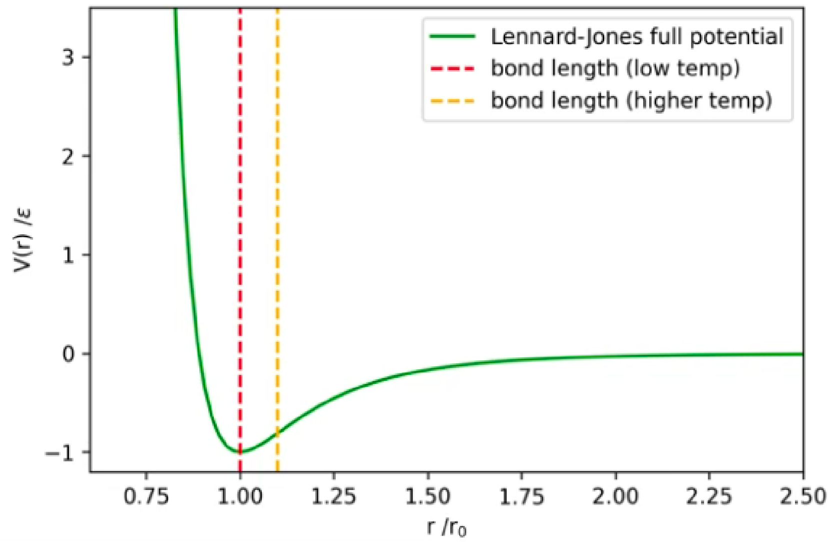

Figure 4.3

Because the LJP is antisymmetric, at higher temperatures the molecules spend less vibrational time in the contracted position and more time in the expanded position, resulting in a larger average bond length. Effectively, they experience more repulsion when they are together than they do attraction when apart, hence a larger bond length on average. This also explains thermal contraction.

We can see from the potential graph that this results in a lower ϵ, which is why we see a change in latent heat at different temperatures.

4 Surface Tension

Consider a (literal) sea of particles. Does a particle surrounded on all sides deep in the liquid experience a lesser or greater potential than a particle on the surface?

The particle deep in the liquid has twice as many nearest neighbours (being surrounded on all sides) so has a greater potential. The particle on the surface will have:

$$V = \frac{n\epsilon}{4}$$

So requires $n\epsilon/4$ amount of work to take a deeper particle to the surface.

We define a quantity called surface tension γ, the work done to move 1m^2 of particles, N to the surface from deeper inside the liquid:

$$\gamma = \left(\frac{n\epsilon}{2} - \frac{n\epsilon}{4}\right) \times N$$

Where N is the number of particles per m^2. It has units of energy over area (force over length).

This allows animals such as small insects who can sit on the surface of the liquid to do so, because allowing the insect to submerge would require an increase in the water's surface area. The work required to generate this new surface area is not achievable by the insects gravitational potential energy.

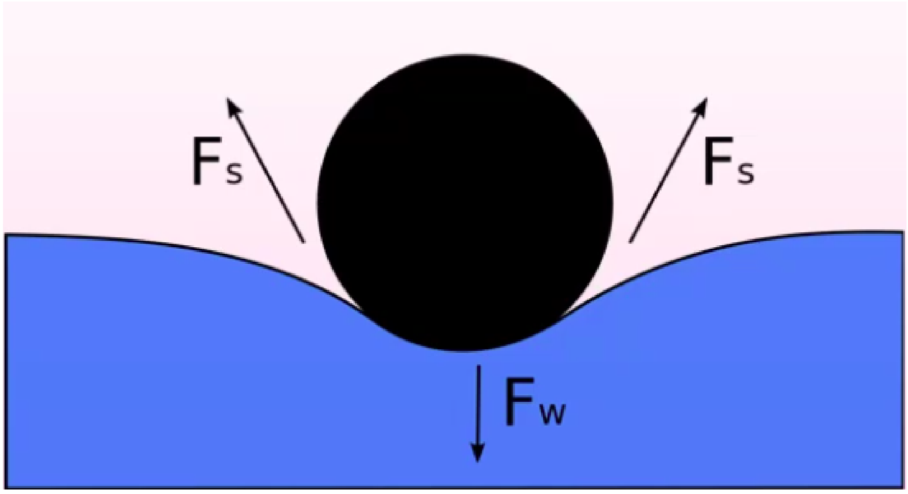

Figure 4.4: Object sinks if $\Sigma F_s < F_w$

What other consequences does this have?

- Given a body of water wants to minimise its surface area (for a constant volume), drops of water will form spheres.

5 Surfaces Revisited

Consider a liquid contained in some beaker. Consider a particle on the surface of the *beaker* (purple):

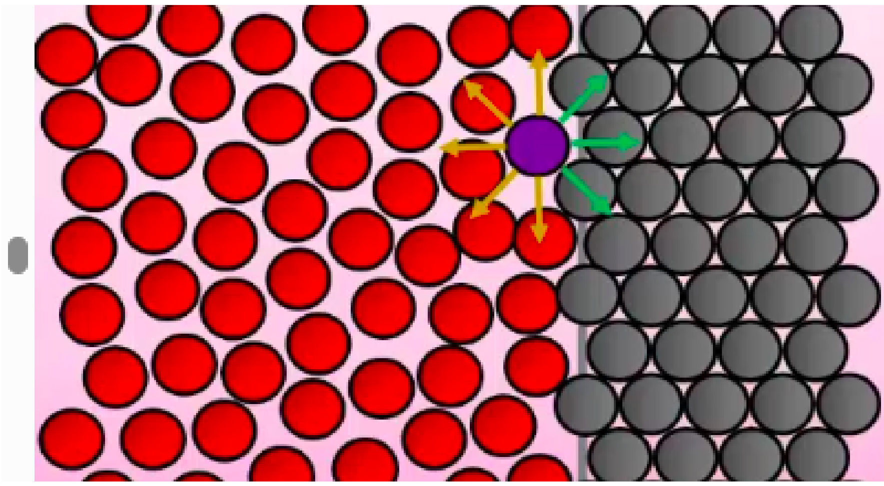

Figure 4.5

There are two forces in effect here, the interatomic forces of the liquid and of the beaker. We call the forces between the liquid itself cohesion, and between the liquid and a foreign body adhesion.

5.1 Wetting

"Wetting" is the ability of a liquid to maintain contact with a solid surface and is inversely proportional to surface tension. Most liquids are "wetter" than water, due to water's high surface tension.

If there are strong adhesive forces, the liquid preferentially sticks to the side of the container, causing the formation of a concave meniscus.

If there are strong cohesive forces, minimisation of surface area causes the formation of a convex meniscus.

6 Capillary Action

If an open tube is placed into contact with the surface of a liquid with strong adhesion, the liquid will rise up the sides of the tube:

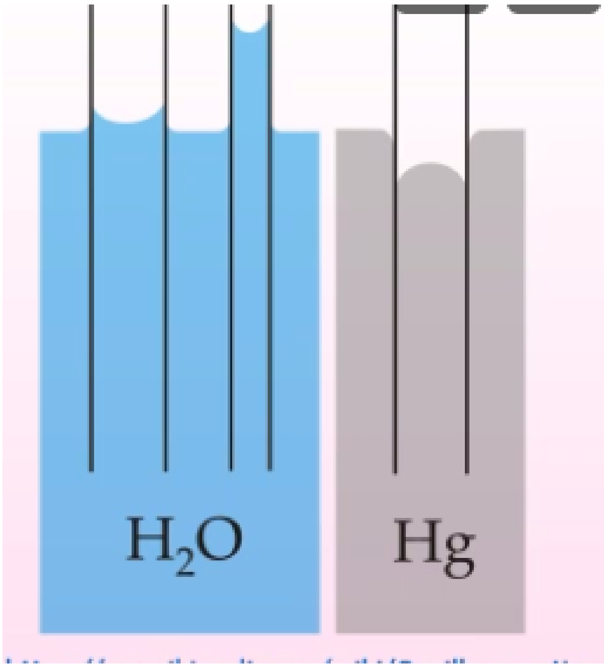

Figure 4.6: Capillary action for water and mercury. Note that mercury, being strongly cohesive, minimises surface area so does not rise up.

We can describe this mathematically in terms of the angle formed by the meniscus, the width of the meniscus and the height reached by the liquid:

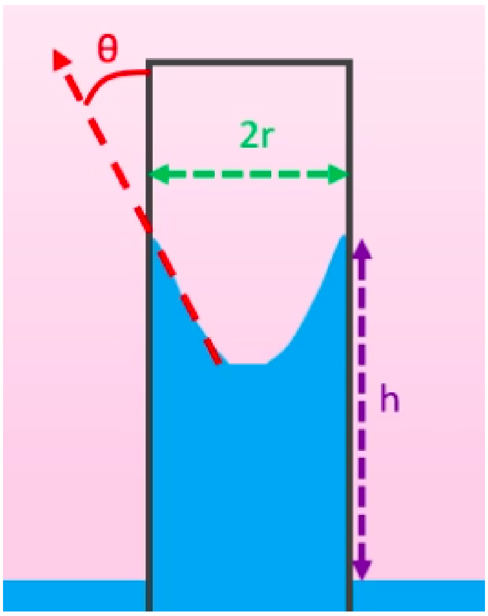

Figure 4.7

There are two competing forces here:

- Surface tension, pulling the liquid up.
- The weight of the liquid, pulling it down.

The weight is given by the volume (if we approximate the meniscus as being small compared to the total volume, hence just a cylinder) and density:

$$F_W = mg = \rho V g = \rho \pi r^2 h g$$

Surface tension has units of force divided by length. We can therefore get the force by the multiplying by the active contact length, which is the circumference of the cylinder. Finally, we only care about the vertical component, so take $\cos\theta$:

$$F_\gamma = \gamma 2\pi r \cos\theta$$

Hence, as equilibrium:

$$F_\gamma = F_W$$

$$2\gamma\pi r \cos\theta = \rho\pi r^2 h g$$

$$2\gamma \cos\theta = \rho g r h$$

This gives:

$$\boxed{h = \frac{2\gamma\cos\theta}{\rho g r}}$$

Tue 03 Feb 2026 12:00

Lecture 5 - Matter V: Stress, Strain and Young's Modulus

1 Definitions

Consider a wire of length L and cross-sectional area A. We apply some force F to the material, causing it to stretch from L to $L + \Delta L$.

Stress is the force applied per unit area on a material, $\mathrm{kgm}^{-1}\mathrm{s}^{-2}$ and strain is the fractional change in length: $\Delta L/L$.

The Young Modulus is the ratio of the two:

$$Y = \frac{\text{stress}}{\text{strain}} = \frac{F/A}{\Delta L/L} = \frac{FL}{\Delta LA}$$

and has units of pressure (Pascals). Different materials have different Young Moduli as inherent material properties.

2 Stress/Strain Graphs

We can consider a plot of stress against strain:

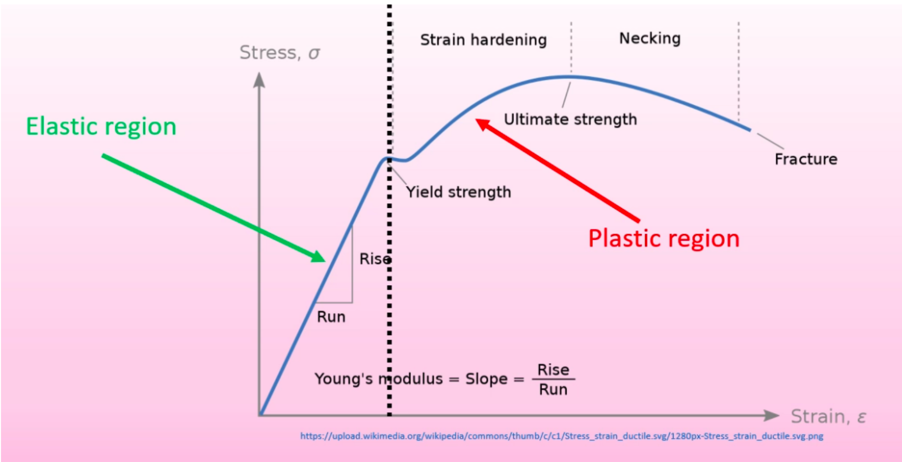

Figure 5.1

The leftmost region is where stress and strain are proportional, i.e:

$$\frac{F}{A} \propto \frac{\Delta L}{L}$$
$$F \propto \Delta L$$

As A and L are constants. Any changes here are reversible, i.e. a rubber band.

$$F = k\Delta L = kx$$

Past a certain extension, we encounter plastic (non-reversible) extension. This is where we've passed the limit of proportionality. There are a few important points here:

- Strain hardening. We apply a force to the object to extend it to its ultimate extension. This means that we can include it in i.e. machinery without worrying about smaller forces regularly encountered causing extensions.
- Necking, where an increase in length causes a decrease in cross-sectional area (as the total volume is held constant)
- Once we get past the strain hardening point, tiny defects cause non-linear necking. A weaker area decreases the cross-sectional area more significantly. This lower A causes the effect to be more significant, and leads to the area becoming thinner and thinner until it fractures.

3 Other Elastic Moduli

3.1 Shear Modulus

For Young's Modulus, we're explicitly considering that the force acts in the direction of the extension. We can have other elastic moduli (still defined as stress over strain) depending on the configuration of the force.

For example, applying a tangential force gives the shear modulus:

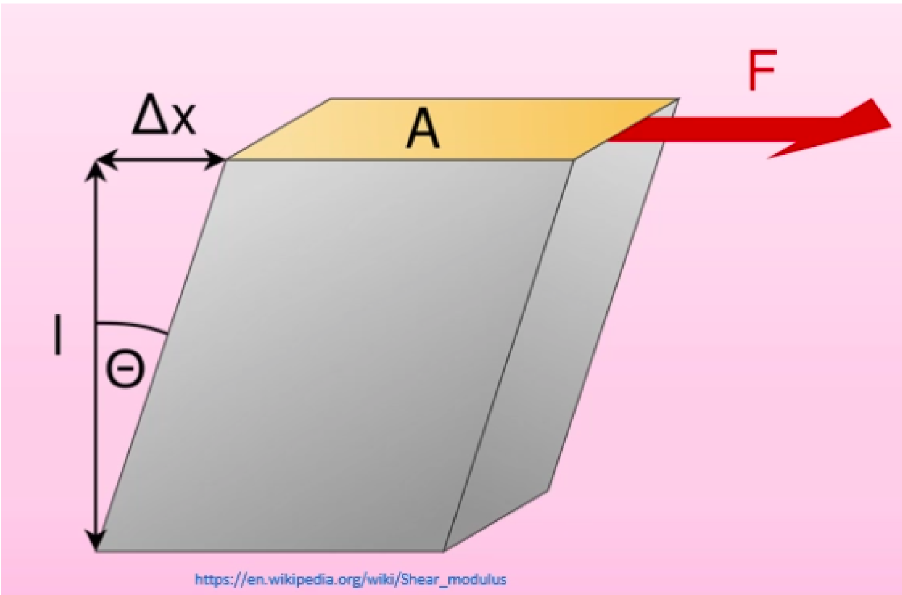

Figure 5.2

The shear stress remains as F/A and the sheer strain is $\Delta x/l$. The sheer modulus is shear stress/shear strain.

3.2 Bulk Modulus

The bulk modulus is for objects deformed under a uniform pressure, for example compressing a liquid from some volume V_0 down to a compressed volume $V_0 - \Delta V$.

The bulk stress is the compressing pressure, F/A and the bulk strain is $\Delta V/V_0$.

So the Bulk Modulus is:

$$\frac{P}{(DeltaV/V_0)} = \frac{PV_0}{\Delta V}$$

Or (using the ideal gas equation we'll encounter later):

$$B = -V_0 \frac{dp}{dV}$$

This is comparable for solids and liquids. We also define the compressibility as the inverse of the bulk modulus.

3.3 Poisson's Ratio

Applying a stress to a force in one axis causes compression in this direction, but expansion in the other:

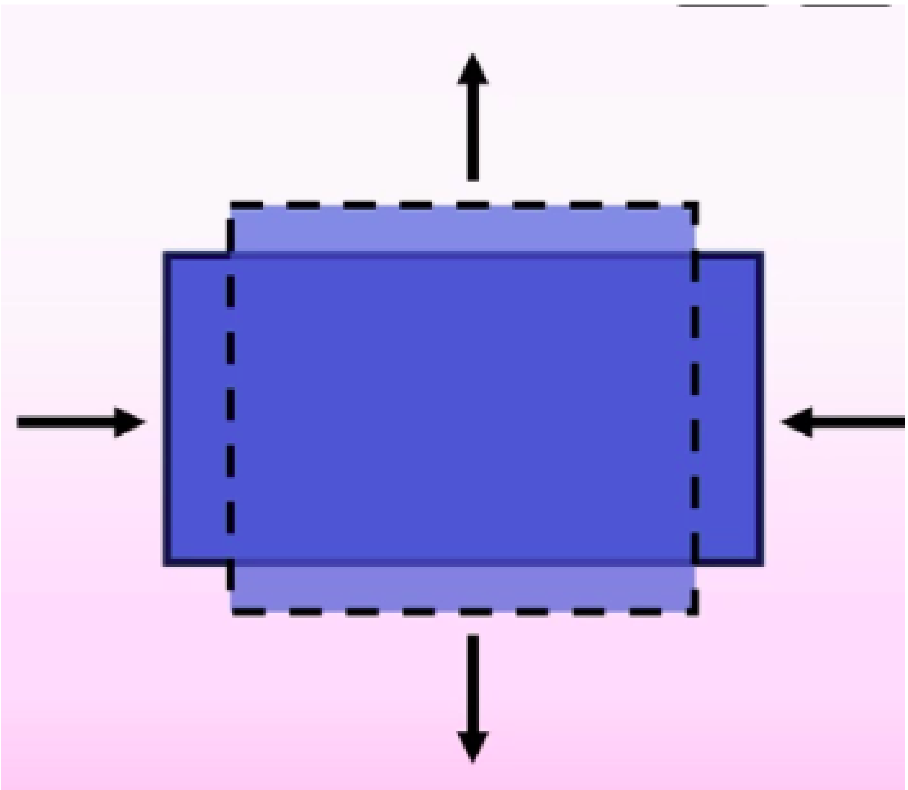

Figure 5.3

This ratio is given by Poisson's Ratio (unitless):

$$\nu = \frac{\text{transverse strain}}{\text{longitudinal strain}}$$

4 Material Values

All of these moduli have varying values as inherent metal properties:

Material	Young's modulus (GPa)	Shear modulus (GPa)	Bulk modulus (GPa)	Poisson's ratio (unitless)
Aluminium	70	30	70	0.33
Brass	91	36	61	0.36
Copper	110	42	140	0.36
Glass	55	23	37	0.22
Iron	190	70	100	0.26
Lead	16	5.6	7.7	0.44
Nickel	210	77	260	0.31
Steel	200	84	160	0.28
Tungsten	360	150	200	0.28

Figure 5.4: Elastic Moduli Values

Note that while the Young and Bulk Modulus values are generally similar, the sheer modulus is often 2-3x smaller.

Thu 05 Feb 2026 13:00

Lecture 6 - Matter VI: LJF Meets Young Modulus and Breaking Materials

1 Young's Modulus Meets Lennard-Jones

On a microscopic scale, applying a linear force extends the distance between neighbouring atoms:

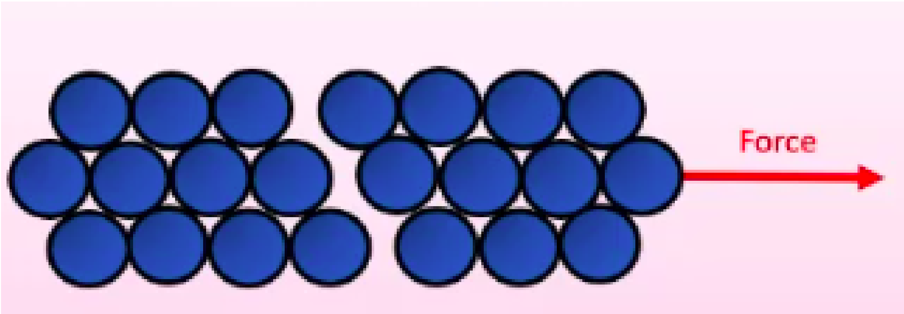

Figure 6.1

The mean interatomic distance changes from $r_0 \to r_0 + dr$ for an infinitesimal force dF. Near the equilibrium point, where the material is in the elastic region, we expect the extension $dr \propto dF$, as in this region $\Delta L \propto F$.

Going back to the Lennard-Jones Force, we have a small region around the equilibrium point:

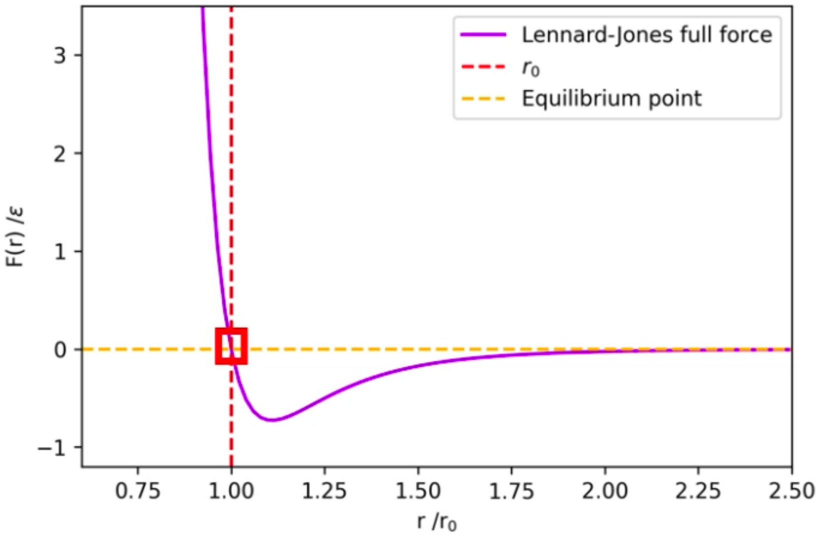

Figure 6.2

If we zoomed into this region, we would see it is approximately linear:

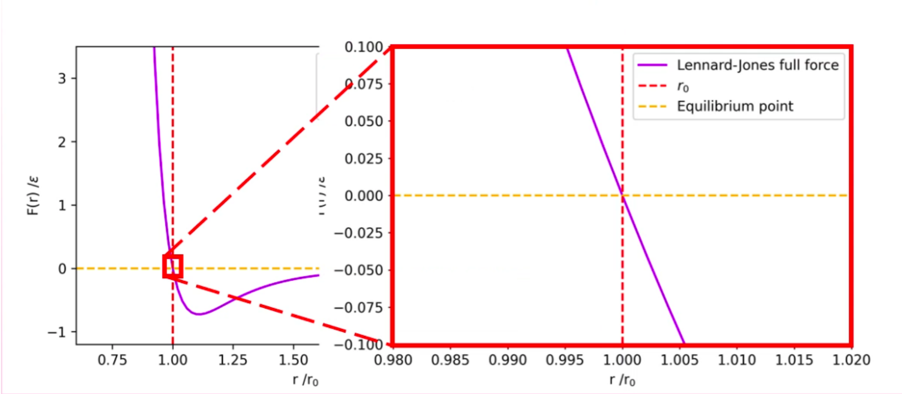

Figure 6.3

Hence, around r_0, dF/dr is constant. Going back to Young's Modulus, we have:

$$Y = \frac{\text{stress}}{\text{strain}} = \frac{F/A}{\delta Y/Y}$$

Here, the force is just dF. The area this force is acting over (microscopically) is actually a square, as the material is a cubic lattice, $A = (r_0 + dr)^2$. Hence stress is:

$$\text{stress} = -\frac{dF}{(r_0 + dr)^2}$$

The negative sign compensates for the fact that the LJF opposes the deformation (is negative under tension), ensuring the stress and strain have the same sign and their ratio is positive.

$$\text{strain} = \frac{dr}{r_0}$$

$$\implies Y = -\frac{\left(\frac{dF}{(r_0+dr)^2}\right)}{\left(\frac{dr}{r_0}\right)}$$

$$= -\frac{dF r_0}{dr(r_0 + dr)^2}$$

We take the quadratic to first order, assuming $dr \ll r_0$:

$$= -\frac{1}{r_0}\frac{dF}{dr}\bigg|_{r=r_0}$$

As force is $-dV/dr$ we can show that:

$$Y = \frac{1}{r_0}\frac{d^2V}{dr^2}\bigg|_{r=r_0}$$

And using the definition:

$$V = \epsilon\left(\frac{r_0^{12}}{r^{12}} - 2\frac{r_0^6}{r^6}\right)$$

Hence:

$$F = -\frac{dV}{dr} = -\epsilon\left(\frac{12r_0^{12}}{r^{13}} - 12\frac{r_0^6}{r^7}\right) = \epsilon\left(12\frac{r_0^6}{r^7} - 12\frac{\epsilon_0^{12}}{r^{13}}\right)$$

$$Y = -\frac{dF}{dr} = \frac{d^2V}{dr^2} = \epsilon\left(\frac{156r_0^{12}}{r^{14}} - \frac{84r_0^6}{r^8}\right)_{r=r_0}$$

$$Y = \frac{156\epsilon}{r_0^2} - \frac{84\epsilon}{r_0^2} = \boxed{\frac{72\epsilon}{r_0^2}}$$

This provides a microscopic definition of the Young's Modulus to complement our previous macroscopic definition.

2 Breaking Materials

Materials break when the attractive force between atoms is less than the force applied to the material. This occurs when:

$$\frac{dF}{dr} = 0 \implies F = F_{\text{max}}$$

If we pull continuously with a force greater than F_{max} we will pull atoms apart and break the material. The maximum definable stress is based on the maximum attractive LJF over the area of a single molecule:

$$\text{Breaking Stress} = \frac{F_{\text{max}}}{r_0^2}$$

2.1 Deriving r at F_{max}

F_{max} occurs where $\frac{dF}{dr} = 0$. Using the earlier result:

$$F(r) = 12\epsilon\left(\frac{r_0^{12}}{r^{13}} - \frac{r_0^6}{r^7}\right)$$

$$\frac{dF}{dr} = 12\epsilon\left(-\frac{13r_0^{12}}{r^{14}} + \frac{7r_0^6}{r^8}\right) = 0$$

$$\frac{7r_0^6}{r^8} = \frac{13r_0^{12}}{r^{14}} \implies 7r^6 = 13r_0^6$$

$$\implies \boxed{r = \sqrt[6]{\frac{13}{7}}\, r_0}$$

2.2 Finding F_{max}

Substituting $r^6 = \frac{13}{7}r_0^6$ into $F(r)$. Simplifying each term:

$$\frac{r_0^{12}}{r^{13}} = \frac{1}{r}\cdot\left(\frac{r_0^6}{r^6}\right)^2 = \frac{1}{r}\cdot\frac{49}{169}, \qquad \frac{r_0^6}{r^7} = \frac{1}{r}\cdot\frac{r_0^6}{r^6} = \frac{1}{r}\cdot\frac{7}{13}$$

$$F_{\text{max}} = \frac{12\epsilon}{r}\left(\frac{49}{169} - \frac{91}{169}\right) = -\frac{504\epsilon}{169\, r}$$

Substituting $r = \left(\frac{13}{7}\right)^{1/6} r_0$:

$$\implies \boxed{|F_{\text{max}}| = \frac{504\epsilon}{169\, r_0}\left(\frac{7}{13}\right)^{1/6} \approx \frac{2.69\,\epsilon}{r_0}}$$

2.3 Breaking Stress

Hence the breaking stress is:

$$BS = \frac{F_{\text{break}}}{A} = \frac{F_{\text{max}}}{r_0^2}$$

Macroscopically and microscopically.

3 Fractures

Say your material has some tiny imperfection of depth d and width $2r$.

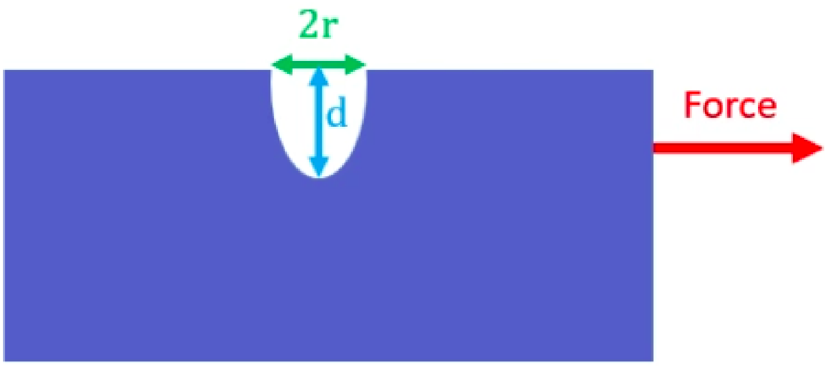

Figure 6.4

This causes stress to increase by a factor of:

$$\left(1 + 2\sqrt{\frac{d}{r}}\right)$$

The breaking force decreases with increasing d, which makes sense as a deeper groove will cause a much smaller cross sectional area and a larger stress.

The breaking force increases with a decreasing r, where grooves on atomic length scales wide causes a huge increase in the stress.

Mon 09 Feb 2026 11:00

Lecture 7 - Thermal Physics I: Introduction to Temperature

We're starting the portion of the module on thermodynamics, looking at temperature, pressure etc and transitions involving these.

1 What is Temperature?

Particle for a single particle isn't well defined for the definition of temperature we want to use. There is two definitions:

- One is statistical, looking at average kinetic energy of particles.
- One is based on entropy and isn't looked at until second year.

We define temperature as *a statistical collection of energies for particles that make up the system for which we are measuring the temperature.* A material will have a range of kinetic energies, some very large, some very small, so we care about a statistical average. Thermalisation happens as a result of these high kinetic energy particles striking lower kinetic energy particles.

> **Temperature:** A statistical collection of energies of the particles that comprise the system in question.

We can express temperature as using the Maxwell-Boltzmann distribution:

$$P(v) = 4\pi\left(\frac{M}{2\pi RT}\right)^{\frac{3}{2}} v^2 \exp\left(\frac{-Mv^2}{2RT}\right)$$

We look at this in more depth when we cover statistical physics later, but for now we care about the key features. If we plot $P(v)$ (the probability of finding a particle at certain velocity) against these velocities for a range of temperatures:

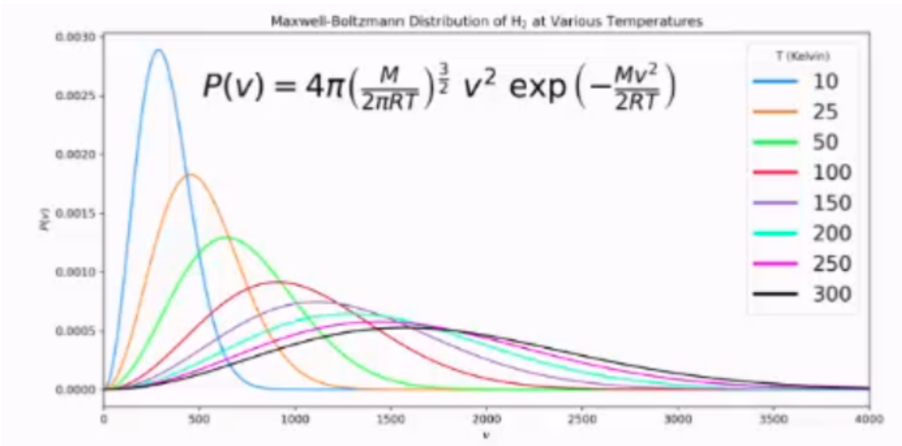

Figure 7.1

This has exponential decay, so theoretically we can find a particle at any velocity within a gas of any temperature. Particles with higher temperatures have a higher peak, so a higher average velocity.

2 Temperature Scales

We can use a number of different temperature scales:

- The Fahrenheit scale is set such that the freezing point of water is 98.6°F, and the boiling point of water at 212°F. It's designed to be based around the range of temperatures that a person could realistically could experience in the core range from 0-200. It's not a useful unit and won't be used in exams.
- Celsius is the standard unit.
- Kelvin is the common unit of thermodynamics. It has the same graduations as degrees celsius but is shifted such that absolute zero is at 0K. Note that Kelvin has no degrees symbol. °C → K can be converted between using:

$$T(\text{K}) = T(^\circ\text{C}) + 273.15$$

3 Thermal Expansion

As a solid is heated, it expands in all directions. Consider a cuboid of height, width, depth W, L, D, we have new dimensions of $D + \Delta W$, $L + \Delta L$, $D + \Delta D$ after a small change in temperature. We can define the linear expansion coefficient (a material property) α_L based on these dimensions and the change in temperature:

$$\alpha_L \Delta T = \frac{\Delta L}{L} = \frac{\Delta W}{W} = \frac{\Delta D}{D}$$

If we consider the area instead, we have the area expansion coefficient:

$$\frac{\Delta A}{A} = \alpha_A \Delta T \approx 2\alpha_L \Delta T$$

The latter approximation can be derived as follows. Consider one dimension, i.e. L:

$$\frac{\Delta}{L} = \alpha_L \Delta T$$
$$\implies \Delta L = L\alpha_L \Delta T$$

So:

$$L_{\text{new}} = L + \Delta L = L\left(1 + \alpha_L \Delta T\right)$$

Pairing this with width to find an area (of a face):

$$W_{\text{new}} = W + \Delta W = W(1 + \alpha_L \Delta T)$$
$$A_{\text{old}} = L \times W$$
$$A_{\text{new}} = L_n \times W_n$$
$$A_n = LW(1 + \alpha_L \Delta T)^2$$
$$= LW\left(1 + 2\alpha_L \Delta T + \alpha_L^2 \Delta T^2\right)$$

For a small change in temperature, the second order term is very small, hence:

$$\frac{\Delta(LW)}{LW} \approx 2\alpha_L \Delta T$$

The linear expansion coefficient has units of K^{-1}.

3.1 Thermal Stress

Consider a rod fixed between two immovable walls. The rod attempts to expand, but is fixed. This provides a thermal stress on the wall. If the rod has cross-sectional area A and Young Modulus Y, the thermal stress is given by:

$$\frac{F}{A} = Y\alpha_L \Delta T$$

This is why bridges etc need to have thermal expansion joints, otherwise they would buckle/break under this thermal stress.

3.2 Lennard Jones and Thermal Expansion

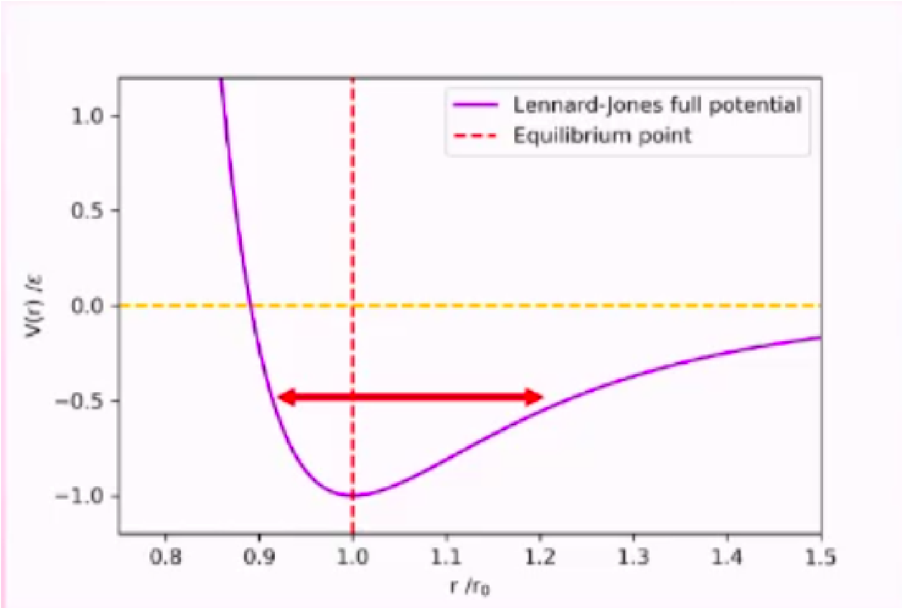

Figure 7.2: LJ Potential

As we increase temperature, the potential decreases and the depth of the potential well decreases. As the LJP is less and less symmetric for increasing r/r_0, this leads to an asymmetric potential where the bonds spend more time on average vibrating in an expanded state than in an equilibrium or contracted state. As energy increases, so does average bond length.

We only see small increases in length as it's an average, some bonds will still be in the equilibrium state or even below it, statistically.

Thu 12 Feb 2026 13:00

Lecture 8 - Thermal Physics II: Thermodynamics Intro

1 Temperature Scales II

In order to measure temperature, we need to define a quantity which directly depends on temperature - ideally linearly.

We can't measure temperature directly, we need this extra intermediary quantity. For example, thermometers use volume and measure the volume of a known quantity of a substance which behaves nicely at known temperatures.

This means that:

$$T + T_0 + k\frac{x - x_0}{x_0}$$

Where T_0 is a calibration point, at which point our quantity has value x_0. For the celsius scale, we use $T_0 = 0°\text{C}$ and we choose a value for the constant k such that when water boils, $T = 100°\text{C}$.

However, $PV = nRT$ says that this is dependant on pressure too which isn't ideal, as for different pressures the relationship between volume and temperature differ and our scale is only consistent for a single pressure....

Instead of using the boiling/freezing point of water at 1ATM has our calibration point, we can use the "triple point" of water. This point only happens at a single pressure/volume, so is more consistent:

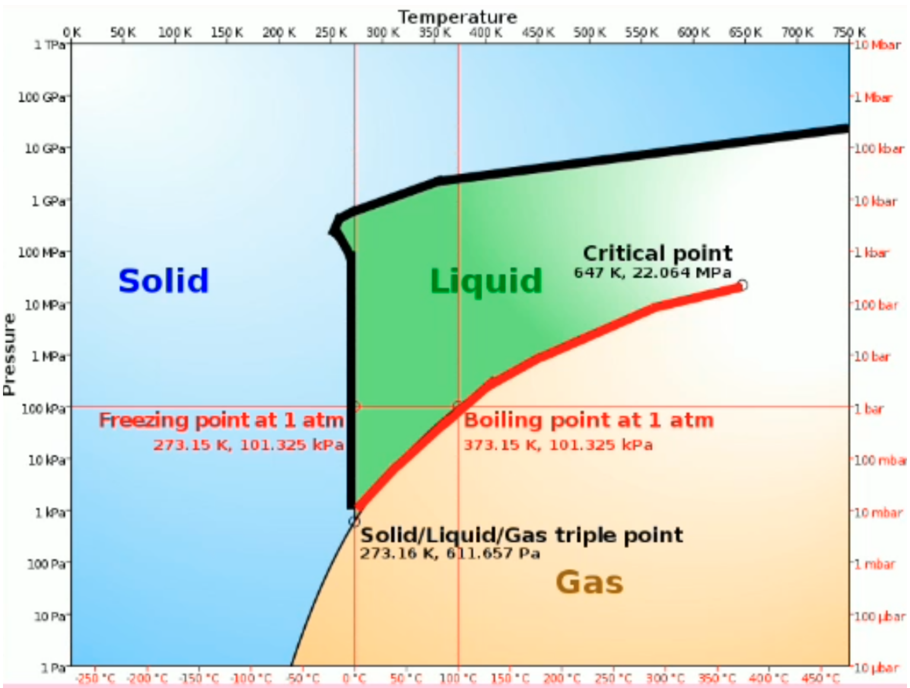

Figure 8.1

Alternatively, we could use the technique employed by the Kelvin scale and use absolute zero as our calibration point. Regardless of pressure/volume/amount of substance (provided its kept constant), they will all reach absolute zero at exactly the same theoretical temperature. We can therefore calibrate our scale off of this point.

2 Thermodynamics

> **Thermodynamics:** The science of the relationship between heat, work, temperature and energy. Thermodynamics broadly deals with the transfer of energy from one place/form to another. Heat is a form of energy corresponding to a definite amount of mechanical work.

2.1 What is heat?

Heat is measure of thermal energy transfer between two systems. This shouldn't be confused with the colloquial understanding of heat i.e. being hot or cold, which is relative.

2.2 Systems

A system is a part of the universe we are investigating or discussing eg a room, bottle, galaxy etc. We often consider the ideal physics case of a closed system where the system doesn't exchange any heat with outside the system however this is not realistic.

We often discuss systems comprised of a container of some kind of fluid which either closed (with a lid) or open (no lid).

We can consider two types of walls:

- Adiabatic walls: No heat transfer possible (does not exist in practice).
- Diathermal walls: Heat transfer is possible through the wall.

2.3 Thermal Equilibrium

A system is in thermal equilibrium if it has a constant energy density throughout, so the system is in internal thermal equilibrium.

Two systems are in thermal equilibrium if there is no net transfer of heat energy between them. There will be some amount of heat transfer between both systems, and thermal equilibrium does not mean that there is no exchange of heat energy between the two. Some of the particles in one part of the system may statistically have higher energy and exchange energy, but there is no *net* transfer.

Consider some gas in a piston with volume, pressure and temperature V_I, P_I, T_I. As we pull out the piston, the volume of the gas changes and we have a new final temperature volume (and as pressure/volume are related to pressure), pressure and temperature V_F, P_F, T_F.

This does not happen instantly and there will be some time required to reach thermal equilibrium (a few seconds) after we have finished pulling out the system.

We can bring a system in contact with another system, i.e. placing a rubber duck (system B) in a bath (system A) or the ocean (system C). Which of these two pairs (A and B vs C and B) will thermalise first? While the duck will reach the temperature of the water faster when in the ocean than the bath, thermal equilibrium requires the entire system to thermalise. Therefore, as the ocean has much higher volume it takes longer for the ocean to entirely thermalise to the newer duck-adjusted temperature.

3 Definitions

- **Isothermal:** A change to a system which takes place at a **constant temperature**.
- **Isobaric:** A change to a system which takes place at a **constant pressure**.
- **Isochoric:** A change to a system which takes place at a **constant volume**.
- **Adiabatic:** A change to a system which takes place **without transfer of heat**. Temperature may change, but heat cannot enter or leave the system.

4 Heat Capacity

What is the relationship between heat energy ΔQ transferred to a system and the corresponding increase in system temperature ΔT? This is given by:

$$\Delta Q = C\Delta T$$

Where C is the heat capacity in J/K, hence:

$$C = \frac{dQ}{dT}$$

C varies with the amount of material, so we additionally define specific and molar heat capacity:

- Specific Heat Capacity J/kgK: $C = mc$
- Molar Heat Capacity $J/molK$: $C = nc$

A higher heat capacity means a smaller increase in temperature for a given heat (i.e. more energy required to raise the temperature of the system by some amount).

To make things more difficult, heat capacity also varies with the system temperature. We also need to specify whether we are measuring isobaric or isochoric, as these will give different values:

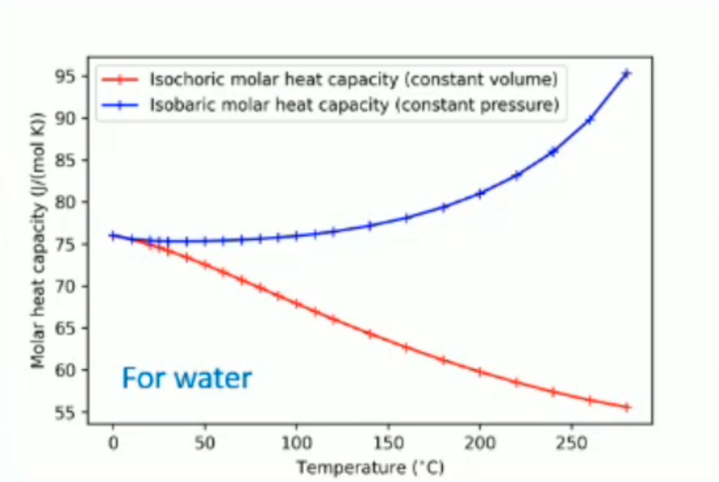

Figure 8.2

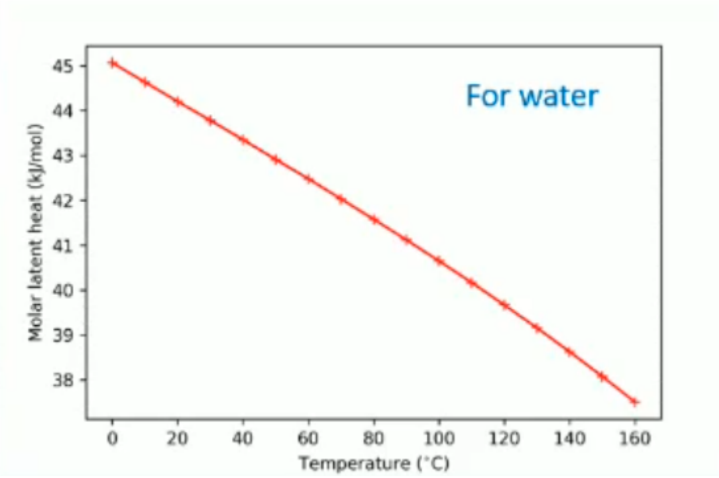

Figure 8.3

Notably, isobaric heat capacity is mostly constant around the standard values of liquid water at standard temperature:

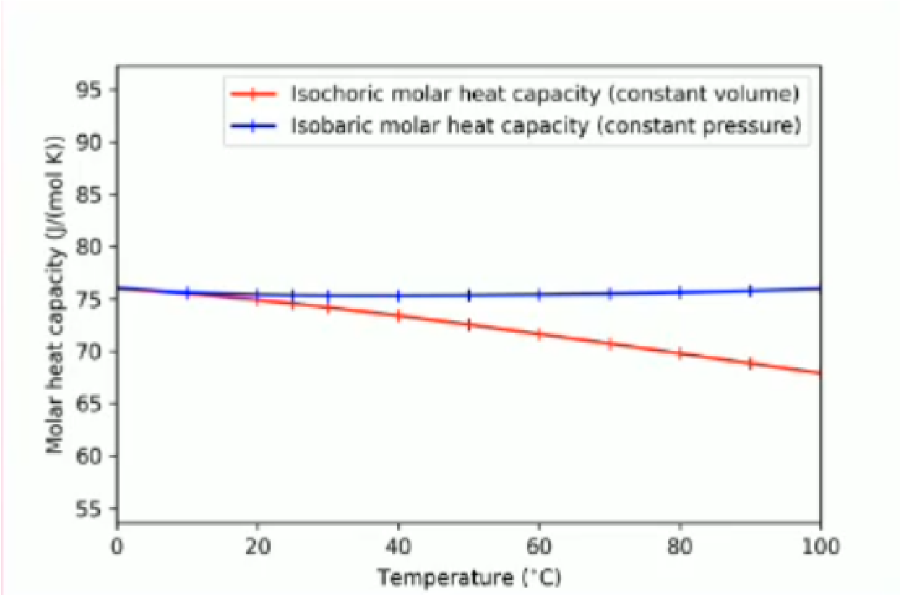

Figure 8.4

It is therefore not unreasonable to treat heat capacity as a constant in some circumstances, provided we state this as an assumption and justify it.

Tue 17 Feb 2026 12:00

Lecture 9 - Thermal Physics III: Laws of Thermodynamics

1 0th Law of Thermodynamics

Consider two blocks next to each other, one with T_h and one with T_c. The blocks with thermalise to their mean $(T_h + T_c)/2$ in the trivial case where the block are identical.

Net heat will flow between thw two systems until they have the same energy density and hence temperature.

If we have three systems: A, B, C. If we know that A and B are in thermal equilibrium, and A and C are in thermal equilibrium, then B and C are also in thermal equilibrium.

It's a fairly trivial axiom, so we denote it the zeroth law. Effectively, all objects in a system in equilibrium share the same temperature.

2 Ideal Gases

An ideal gas is defined as a collection of molecules (or atoms, if monatomic) that are non-interacting with each other (no interatomic forces) and collide elastically with each other.
The internal energy of the gas is dependant on the velocities of the molecules, and hence on the temperature, and not on pressure or volume.

Boyle's Law: "The absolute pressure exerted by a given mass of an ideal gas is inversely proportional to the volume is occupies, if the temperature and amount of gas remain unchanged within a closed system."

Effectively:

$$P \propto \frac{1}{V}$$

Charles' Law: "When the pressure of a sample of an ideal gas is held constant, the Kelvin temperature and volume will be in direct proportion."

Effectively:

$$T \propto V$$

Ideal Gas Law: Since $P \propto \frac{1}{V}$ and $T \propto V$, we have: $PV = kT$, where k varies with context, for example when considering moles:

$$\boxed{PV = nRT}$$

Where n is the number of moles and $R = 8.314\text{Jmol}^{-1}\text{K}^{-1}$ is the gas constant.
Or for molecules:

$$\boxed{PV = Nk_bT}$$

Where $N = N_a n$ is the number of molecules, and $k_b = R/N_a$ is the Boltzmann Constant.

This is called an equation of state and allows us to describe the gas' state macroscopically. We generally express this on a P/V diagram, where each point on the plot represents a specific gas state. If we keep the amount of gas present constant, we can use any two of the variables to determine the third.

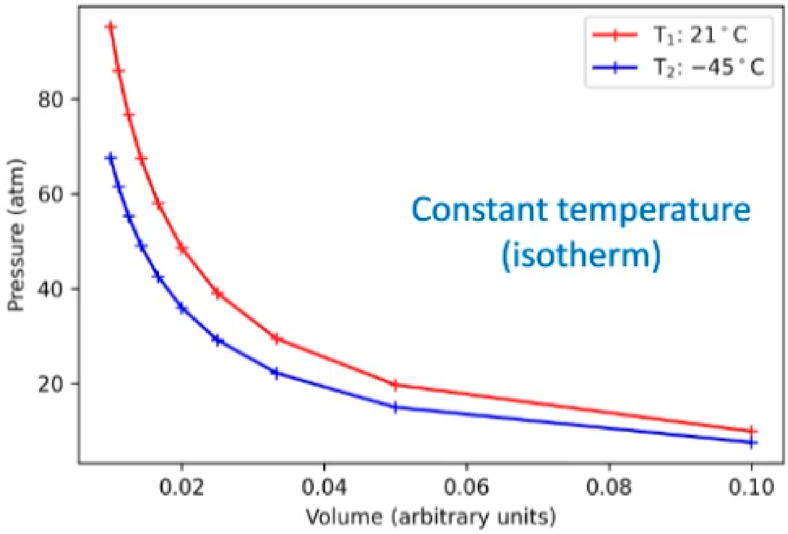

Figure 9.1

2.1 Joule's Second Law

Consider a system with a box divided in two. An ideal gas is contained in the leftmost half, and the rightmost half is a vacuum. We separate the two with a divider constituting an impermeable membrane.

We remove the membrane laterally, doing no work on the gas. Joule observed that the gas stayed at the same temperature as it diffused into the vacuum.

Consider the internal energy of the gas, denoted U, assuming that U is a function of two of the state variables. We know the temperature did not change, and the volume did, so $U(T, V)$.

We would expect:

$$dU = \frac{\partial U}{\partial T}dT + \frac{\partial U}{\partial V}dV$$

Temperature was observed experimentally to not change, hence $dT = 0$, so:

$$dU = 0 + \frac{\partial U}{\partial V}dV$$

Volume did change, so $dV \neq 0$. We did no work on the gas, as the divider was removed laterally, hence there was no change in internal energy, so:

$$dU = \frac{dU}{dV}dV = 0$$

And since $dU = 0$, we must have:

$$\frac{dU}{dV} \neq 0$$

This means that there is no dependence on volume for internal energy, hence U is dependent only on T, as found by Joule. This means that the gas Joule chose was well approximated by an ideal gas.

This is easier at higher temperatures, as a high temperature leads to high kinetic energies, so the interatomic forces becomes less significant and easier to disregard in reality.

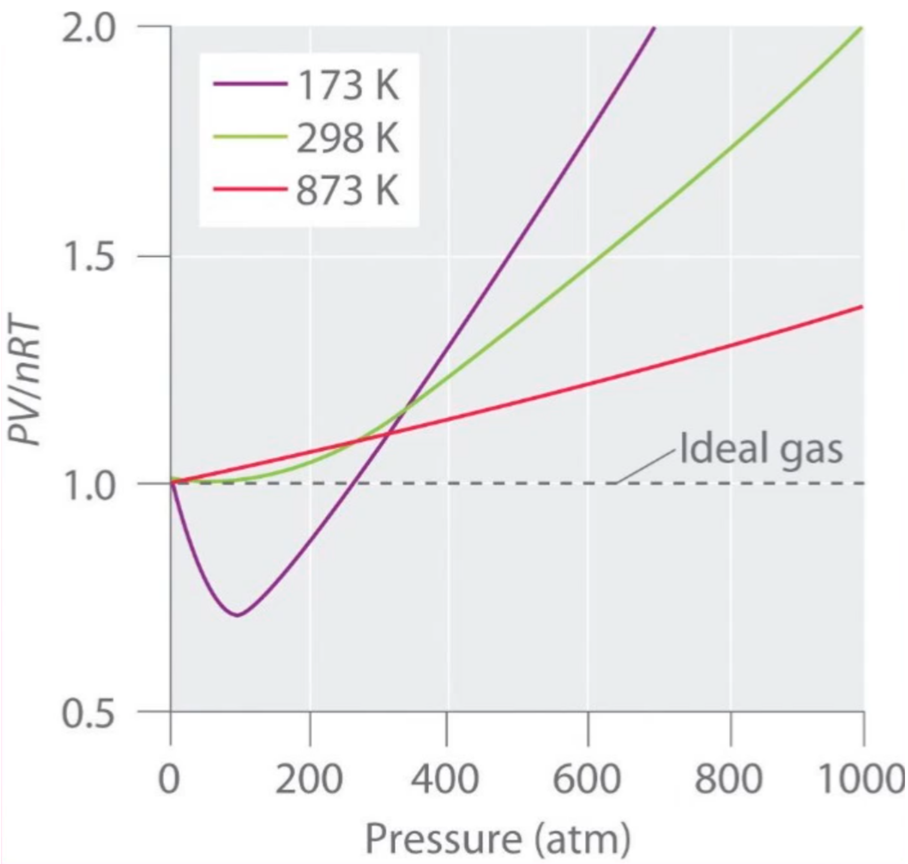

Figure 9.2

We see here that it the ideal gas law becomes a better approximation has we increase temperature, and a worse and worse approximation as pressure increases.

From a LJP perspective, increasing the pressure decreases the average distance between gas molecules. This means that the potential between gas molecules is no longer negligible, and the assumption of zero potential no longer holds.

If we increase pressure even further, the force becomes repulsive and PV/nRT returns to being positive. At higher temperatures, this little dip isn't observed as a higher kinetic energy makes any interatomic forces relatively more negligible.

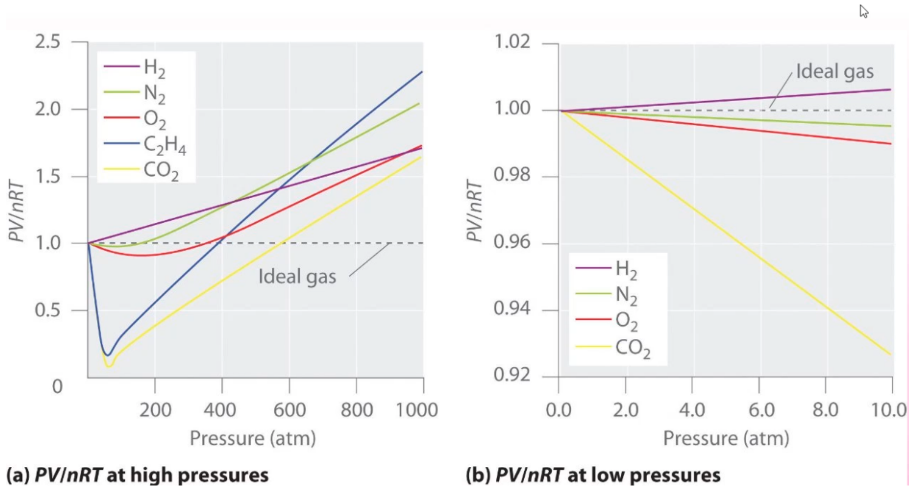

Figure 9.3

At low pressure or high temperature, the ideal gas law is a good assumption as:

- At low pressure, interatomic spacing is large enough to disregard interatomic forces.

- At high temperature, kinetic energy is large enough to comparatively disregard interatomic forces.

We deal solely with Maxwell-Boltzmann gases in this source that follow classical laws. We also have Fermi and Bose gases (made entirely of fermions and bosons respectively), but they're quantum mechanical, exotic and outside of this course.

2.2 Changing Energies

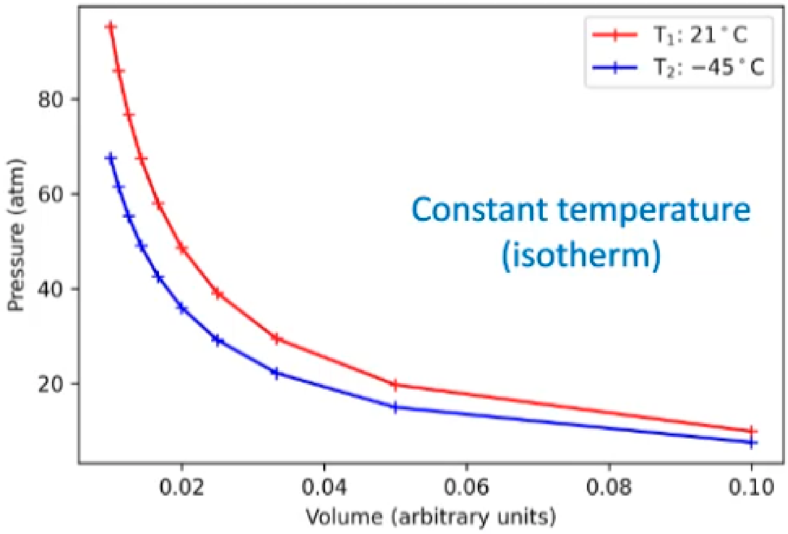

Figure 9.4

Say we want to go from T_1 to T_2. We need to change the internal energy of the gas somehow. We have two general ways to change this:

- Add heat to the system by transferring heat to the gas at a constant volume.
- Do some work on the gas, i.e. by crushing it and reducing volume.

Consider a piston of area A, applying constant force F with extension ΔL. The pressure on the gas is $P = F/A$. The work done on the gas is:

$$\Delta W = F\Delta L$$
$$\implies \Delta W = P(A\Delta L)$$
$$\implies \Delta W = P\Delta V$$

Moving from infinitesimal Delta values:

$$W_{\text{on gas}} = \int dW = \int P\,dV$$

In the opposite case considering work done by the gas, conservation of energy says it must be oppositely signed:

$$W_{\text{by gas}} = -\int P\,dV$$

2.3 Mass Drop Experiment

The falling mass spins the paddles in the water, the work as potential energy is converted to kinetic energy raising the temperature of the water.

This tells us that mechanical work done and heat energy must be related in some manner.

3 1st Law of Thermodynamics

The change in internal energy of a system, ΔU is increased with increasing heat transfer into the system, Q_{in} and work done on the system, W_{on}:

$$\Delta U = Q_{\text{in}} + W_{\text{on}}$$

This is just conservation of energy.

Thu 19 Feb 2026 13:00

Lecture 10 - Thermal Physics IV: Thermodynamic Transitions

1 Thermodynamic Transitions

1.1 Isothermal (Fixed Temperature)

This may look like:

- A curve on a PV plot. As $PV = nRT$, a PV plot shows a $1/x$ relationship for constant T.
- A vertical straight line on a PT plot.
- A vertical straight line on a VT plot.

If the temperature change is 0, ΔU (which we derived last time only depends on temperature) follows:

$$\Delta T = \Delta U = 0$$

Hence:

$$0 = Q_{\text{in}} + W_{\text{on}}$$

1.2 Isochoric (Fixed Volume)

This may look like:

- A vertical straight line on a PV plot.
- A proportional relationship on a PT plot.
- A horizontal straight line on a VT plot.

For an isochoric transition, $\Delta U = Q_i n$, as:

$$W_{\text{on}} = (-) \int P dV$$

And as $dV = 0$, $W_{\text{on}} = 0$

1.3 Isobaric (Fixed Pressure)

This may look like:

- A horizontal straight line on a PV plot.
- A horizontal straight line on a PT plot.
- A proportional relationship on a VT plot.

For an isobaric transition, P is just a constant, so the integral:

$$W_{\text{on}} = \int_{V_1}^{V_2} P\, dV = [PV]_{v_1}^{v_2}$$

Where P is constant, which is nice and easy. Generally:

$$W_{\text{on}} = -\int PdV = (-)P\Delta V$$

$$\Delta U = \Delta Q - P\Delta V$$

2 Heat Capacities of Ideal Gases

Recall that:

$$\Delta Q = C\Delta T \implies C = \frac{dQ}{dT}$$

At a fixed volume:

$$C_V = \left(\frac{dQ_{\text{in}}}{dT}\right)_T$$

And at a fixed pressure:

$$C_P = \left(\frac{dQ_{\text{in}}}{dT}\right)_P$$

Mayer's Relation says that:

$$C_P = C_V + nR$$

This is an examinable derivation.

2.1 Deriving Mayer's Relation

Proof. Consider a gas piston. We have state variables V, P, T, and N is constant as the piston is sealed. The first law of thermodynamics says:

$$\Delta U = Q_{\text{in}} + W_{\text{on}}$$

We can consider this for both a constant volume or a constant pressure. For a constant volume:

$$\Delta U = \Delta Q_1$$

And for a constant pressure:

$$\Delta U = \Delta Q_2 - P\Delta V$$

Noting that the two Qs are different as we have two different transitions. We can create a slightly different expression for constant volume by multiplying by $\Delta T/\Delta T$:

$$\Delta U = \frac{\Delta Q_1}{\Delta T} \times \Delta_T = C_V \Delta T$$

And for constant pressure:

$$\Delta Q_2 = \Delta U + P\Delta V = \frac{\Delta Q_2}{\Delta T}\Delta T = C_P \Delta T$$

Hence:

$$C_P \Delta T = \Delta U + P\Delta V$$
$$C_V \Delta T = \Delta U$$

Setting equal to each other based on ΔU:

$$\Delta U = \boxed{C_V \Delta T = C_P \Delta T - P\Delta V}$$

$$C_P \Delta T - C_V \Delta T = P\Delta V$$

$$\Delta T(C_P - C_V) = P\Delta V$$

$$\implies C_P - C_v = \frac{\Delta V}{\Delta T} \times P \qquad \text{NB: } P \text{ is still constant.}$$

Using $PV = nRT$, for constant pressure we have $\frac{P\Delta V}{\Delta T} = nR$. Hence:

$$C_P - C_V = nR$$

$$C_P = C_V + nR$$

$\square$

It is clear that $C_P > C_V$.

Note: We tend to use heat capacity at a constant volume unless specified.

3 Back to Thermodynamic Transitions

3.1 Adiabatic Transitions (No Heat Transfer)

These seem quite a bit like isothermal transitions, but look different on a PV plot:

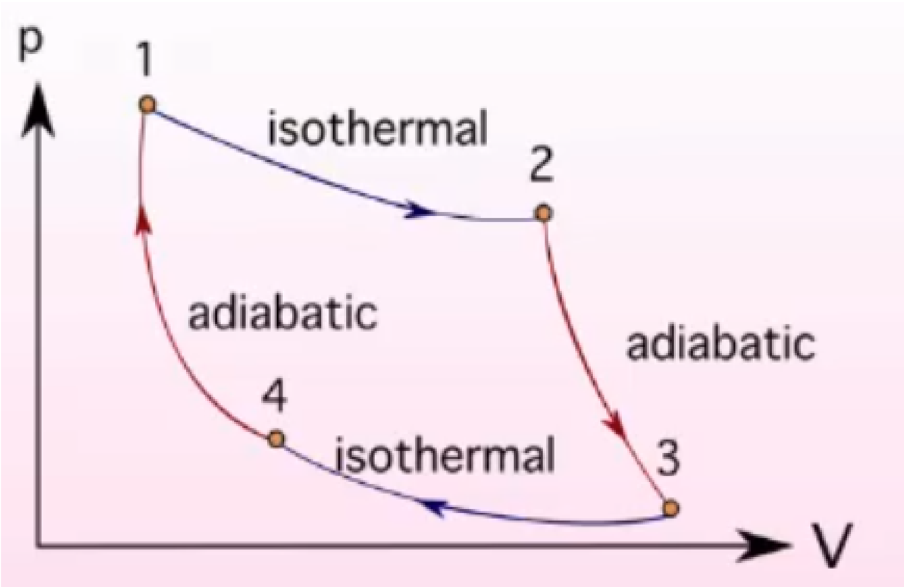

Figure 10.1: Differences in transitions on a PV Plot

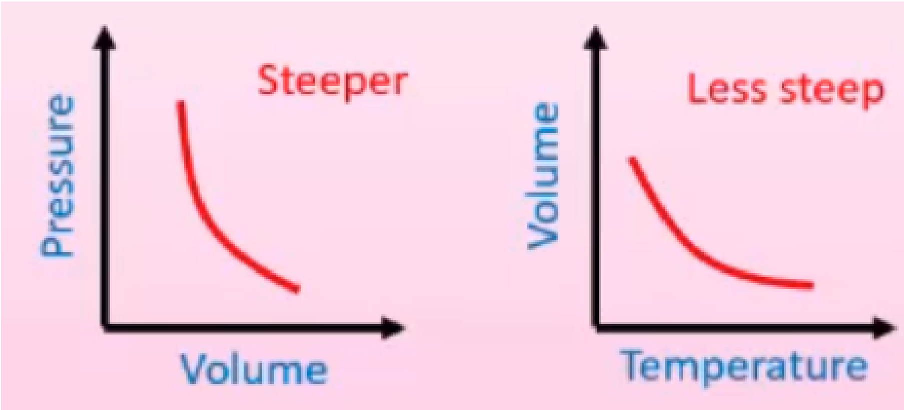

Figure 10.2: Adiabatic Transitions on a PV and VT Plot

For isothermal transitions, $PV =$ const.

For adiabatic transitions, $PV^\gamma =$ const where $\gamma = \frac{C_P}{C_V}$

In a PV plane, adiabatic processes appear steeper than isothermal processes and adiabatic processing are steeper in the PV plane than the VT plane.

For an adiabatic transition, we can reduce the first law of thermodynamics like so:

$$\Delta U = Q_{\text{in}} + W_{\text{on}}$$

$Q_{\text{in}} = 0$ as there is no heat transfer, hence:

$$\Delta U = W_{\text{on}}$$

3.2 Derivations

Proof. For adiabatic transitions, where $Q = 0$.

The First Law becomes $\Delta U = Q_{\text{in}} + W_{\text{on}}$. Since $Q_{\text{in}} = 0$:

$$dU = 0 + dW_{\text{on}}$$

And assuming work is being done *on* the gas, the sign becomes negative:

$$dU = -PdV$$
$$\frac{dUdT}{dT} = -PdV$$

We slightly abuse the definition of heat capacity here, and this will be elaborated on in a later lecture:

$$C_V dT = -PdV$$

Using $PV = nRT$, we have: $d(PV) = nRdT$:

$$nRdT = PdV + VdP$$
$$dT = \frac{PdV + VdP}{nR}$$
$$C_V dT = -PdV$$
$$0 = C_V \frac{PdV + VdP}{nR} + PdV$$
$$0 = C_V(PdV + VdP) + nRpdV$$
$$0 = PdV(C_v + nR) + C_V VdP$$
$$0 = PdV(C_P) + VdP(C_V)$$
$$0 = \frac{C_P}{C_V}PdV + VdP$$

Let $\gamma = \frac{C_P}{C_V}$, and dividing through by pressure and volume:

$$\gamma\frac{dV}{V} + \frac{dP}{Ps}$$
$$\int \gamma\frac{dV}{V} = -\int \frac{dP}{P}$$
$$\gamma \ln V + c_1 = -\ln(P) + c_2$$

Finally:

$$\gamma \ln(v) + \ln(p) = \text{const}$$
$$\ln(PV^\gamma) = \text{const}$$
$$PV^\gamma = e^{\text{const}} = \text{another constant}$$

Hence:

$$PV^\gamma = \text{const}$$

As required! □

Note that in a TV plane, we instead have:

$$TV^{\gamma-1} = \text{const}$$

Tue 24 Feb 2026 12:00

Lecture 11 - Thermal Physics V: PV Planes and Transitions

Note RE: Last Lecture
To avoid confusion, the Physics convention is:

$$W_{\text{by}} = \int PdV$$

$$W_{\text{on}} = -\int PdV$$

1 PV Diagrams

We can move between points on a PV plane through processes which usually involve some kind of heat transfer and work. For example:

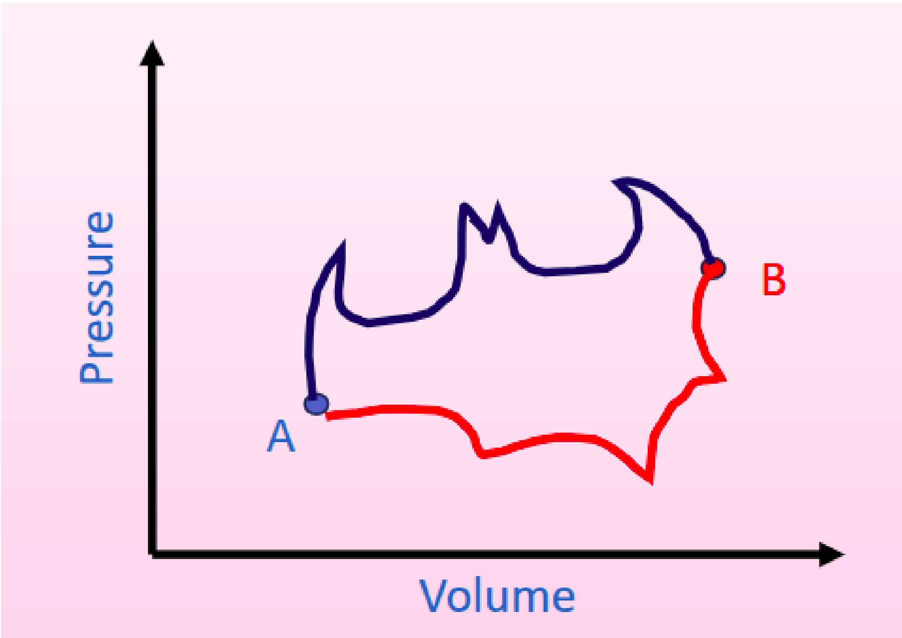

Figure 11.1

The two transitions from $A \to B$ are processes, and the nature of the process depends on the shape of the transition on the PV plot.

- Since the two possible transitions between A and B share the same start and end point, the change in internal energy U_{AB} is the same for both. Since we go from T_A to T_B in both cases (as U depends on T, which in turn depends on P, V so for the same P, V we have the same T), both paths have the same start/end internal energy, so the same change between them.
- The work done by the gas in the two paths is not the same, however. W_{by} is larger for the blue path as W is an integral of $\int PdV$, which represents the area under the curve. The blue line is further up the pressure axis, so the integral is larger and more work is done.

- What about heat input to the gas, Q_{in}? Since the First Law says $\Delta U = Q_{\text{in}} + W_{\text{on}} \implies \Delta U = Q_{\text{in}} - W_{\text{by}}$. ΔU is constant, and $-W_{\text{by}}$ has a larger magnitude for blue, so must Q_{in} to keep ΔU constant.

These transitions are called a *PV Cycle* as we can start at A, go to B and end back up at A.

Consider a simple example with straight lines:

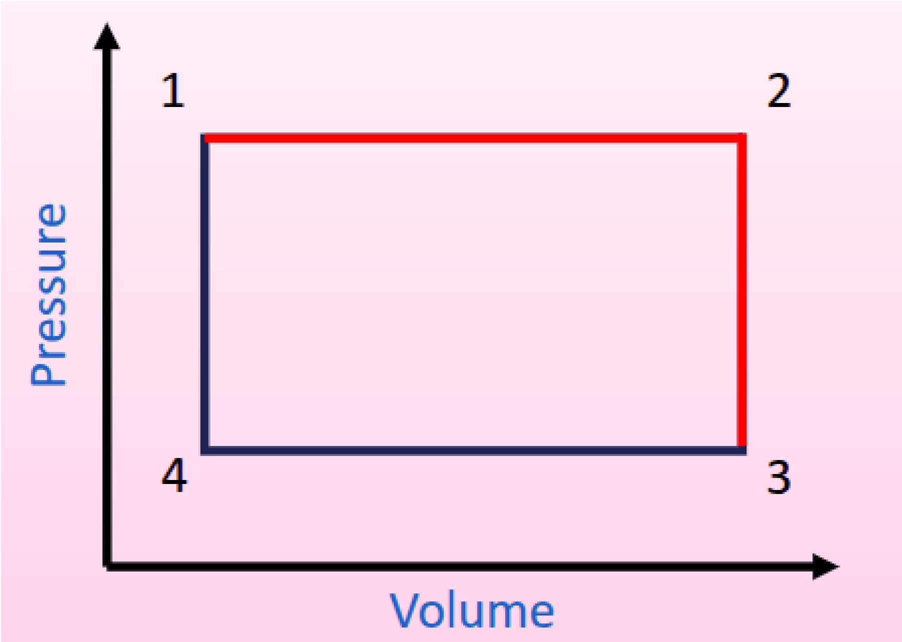

Figure 11.2

Where $1 \rightarrow 2$ and $3 \rightarrow 4$ are isochoric processes and $2 \rightarrow 3$ and $4 \rightarrow 1$ are isobaric. We can also look at isothermal and adiabatic processes:

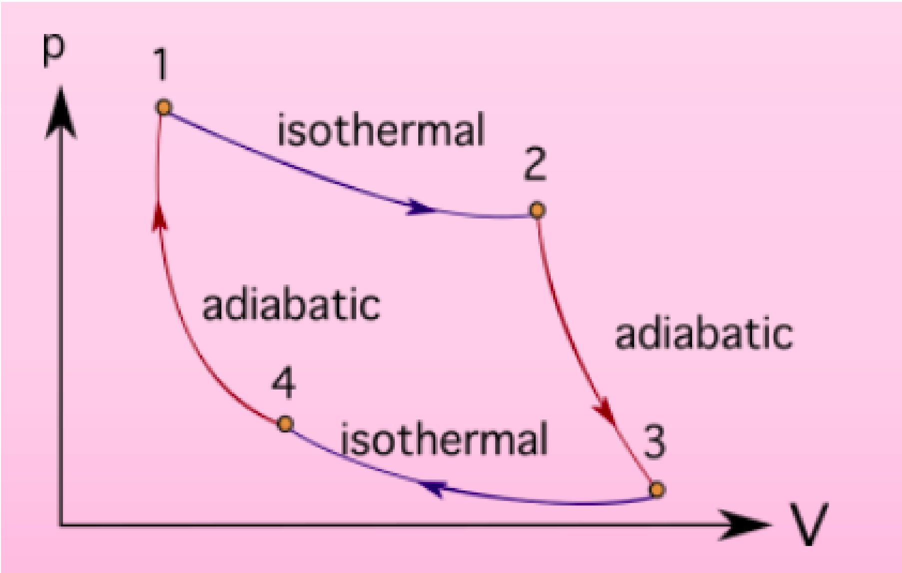

Figure 11.3

A cycle must follow these rules:

1. $\Delta U = 0$, for a full cycle.
2. Total work done is given by the area contained by the cycle.
3. Clockwise cycle for positive work, anticlockwise for negative work.

4. $Q_{\text{in}} + W_{\text{on}} = 0$ as $\Delta U = 0$.

2 Otto Cycle

This describes a petrol internal combustion engine:

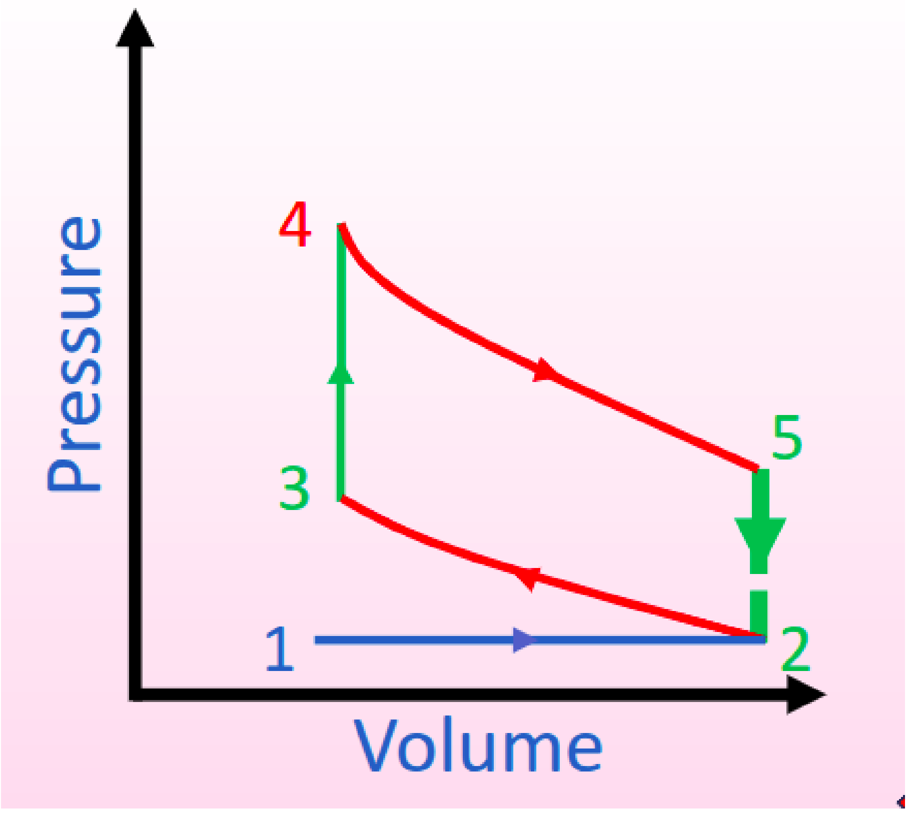

Figure 11.4: Blue: Isobaric, Green: Isochoric, Red: Adiabatic

This has the following components:

1. $1 \rightarrow 2$: Intake stroke: Isobaric introduction of air and fuel to the system.
2. $2 \rightarrow 3$: Adiabatic compression of the fluid as the piston compresses it.
3. $3 \rightarrow 4$: Isochoric introduction of heat energy to the system (ignition of the fluid mixture).
4. $4 \rightarrow 5$: Adiabatic expansion of the fluid mixture as the gas does work on the piston (pushing it down).
5. $5 \rightarrow 2$: Isochoric expulsion of heat from the system.
6. $2 \rightarrow 1$: Mass of air and exhaust gases released at a constant pressure.

3 Diesel Cycle

Here, the ignition and combustion process is performed at a constant pressure where work is done by the gas:

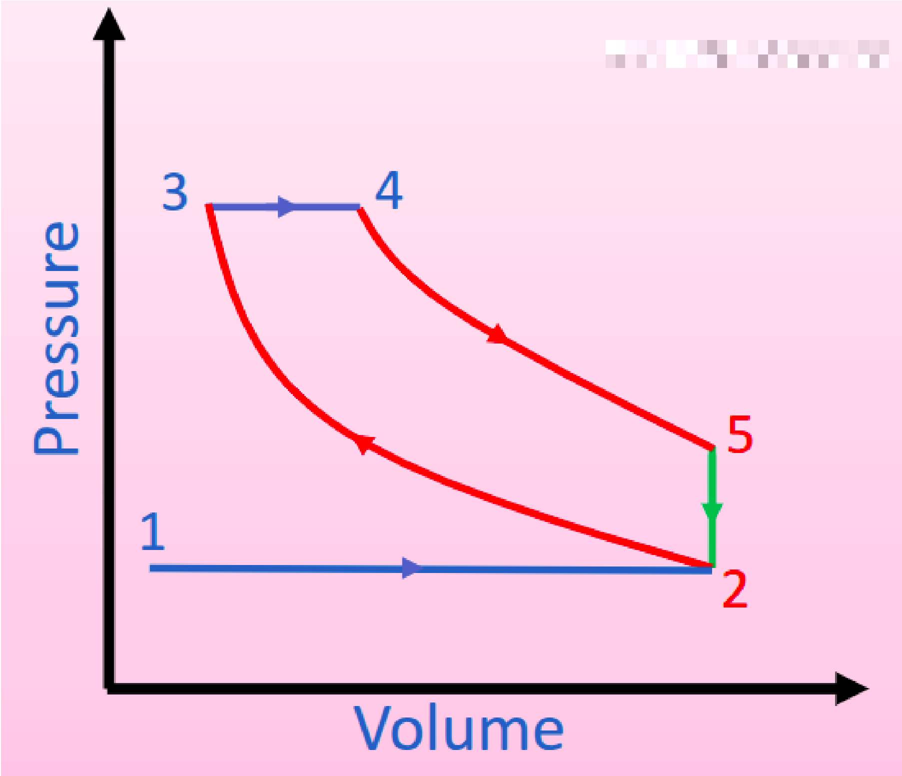

Figure 11.5: Blue: Isobaric, Green: Isochoric, Red: Adiabatic

In both cases, heat is introduced (by combustion) from 3 → 4, denoted Q_1 and heat is lost from 5 → 2 when heat is expelled from the system. The useful work done is $Q_1 - Q_2$ and is given by the area enclosed by the loop.

3.1 Example

Consider the adiabatic gas compression stroke in a diesel engine. (This is the adiabatic stroke where pressure is increasing/volume is decreasing, i.e. from 2 → 3.)

10^{-3}m^3 of nitrogen gas is initially at atmospheric pressure and 298K and is compressed to 1/15th of its original volume.

You may assume that nitrogen is an ideal diatomic gas with $\gamma = 1.4$ and $C_V = 20.85J/Kmol$.

Calculate:

1. The final pressure.
2. The final temperature.
3. The number of moles in the gas.
4. The change in internal energy.
5. The work done by the gas.

Part I
Atmospheric pressure is $101325 \approx 10^5$Pa. As this is an adiabatic transition, $PV^\gamma = \text{const}$. Hence:

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$\implies P_2 = P_1 \left(\frac{V_1}{V_2}\right)^\gamma$$

$$\implies P_2 = P_1 \left(\frac{V_1}{(V_1/15)}\right)^\gamma$$

$$\implies P_2 = P_1 (15)^\gamma = 4.5\text{MPa}$$

Part II
We can use either $PV = nRT$ twice, using the initial conditions to determine n or we can use:

$$T_1 V_1^{(\gamma-1)} = T_2 V_2^{(\gamma-1)}$$

Using the same process gives:

$$T_2 = T_1(15)^{(\gamma-1)} = 880\text{K}$$

Part III
Since we know P_1, V_1 or P_2, V_2 we can use $PV = nRT$, as mentioned above. It would be a good sanity check to do both and compare the values.

Part IV
It's an adiabatic transition, we can related $C_V = \frac{\partial U}{\partial T}$.

Integrating both sides gives:

$$\int dT = \int \frac{\partial U}{\partial T}$$

Thu 26 Feb 2026 13:00

Lecture 12 - Thermal Physics VI: Heat Transfer

1 Thermal Transport

> **Newton's Law of Cooling:** "The rate of heat loss of a body is directly proportional to the difference in the temperatures between the body and the environment"

There are three means of thermal transport:

- **Conduction:** Heat is transmitted directly from one material to an adjoining material (or across parts of the same material) if there is a temperature difference between the two. There is no movement of the material.
- **Convection:** Movement of particles through a substances (often particles that comprise a fluid) that transport their heat energy along with them.
  - The Archimedes principle describes buoyancy based on the material's density ρ, g and the volume of the the fluid displaced by the object, V:
  $$F_b = \rho g V$$
  - In convection, heating a particle causes an increase in bond size (by the LJP), hence the density decreases. This causes it to displace gas of a larger density, hence it rises.
- **Radiation:** Emission of electromagnetic radiation by all bodies which have heat (i.e. all bodies full stop), depending on their temperature.

1.1 Heat Baths

A heat bath is an object with a sufficiently large heat capacity, C such that a large change in its heat energy will result in a negligible change in temperature.

2 Conduction

Suppose we have a small uniform rod of length L and cross-sectional area A linking two heat baths. The rod is conducting and the heat baths have temperature $T_1 < T_2$.

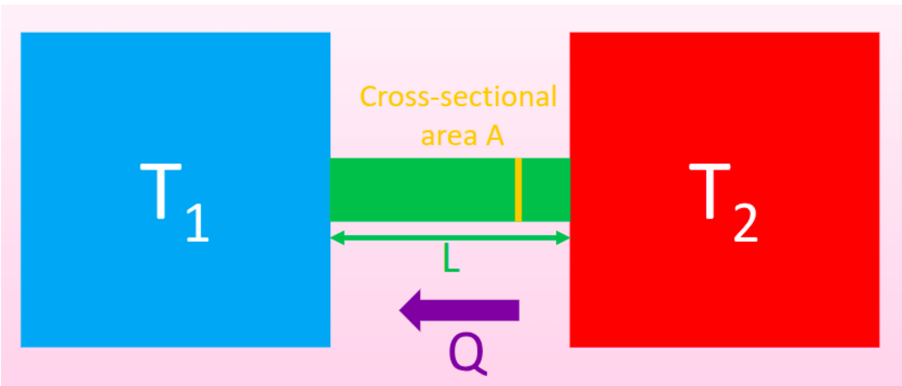

Figure 12.1

Eventually, we reach a "steady state" where all points on the road have a $\dot{Q}$ which does not change with time.

$$\frac{dQ}{dt} = \dot{Q} = \text{constant}, \qquad \forall \text{x along rod (in steady state)}$$

What does this value of $\dot{Q}$ depend on? For a small section dx of the rod:

- Temperature change of the small section, dT.
- Cross-sectional area of the rod A.
- The "thermal conductivity" of the rod, κ[1]. It has units $\text{Wm}^{-1}\text{K}^{-1}$.

Putting these together gives us Fourier's Law:

$$\dot{Q} = -\kappa A \frac{dT}{dX}$$

As T_2 is hotter, the flow of heat is from right to left. Therefore until thermalisation occurs, temperature increases from left to right across the rod ($\frac{dT}{dx} > 0$). However, the heat flow is in the opposite direction from right to left, hence the negative sign in the formula for $\dot{Q}$.

We want to show that $\frac{dT}{dx} = \frac{T_2 - T_1}{L}$:

$$\begin{aligned}
\dot{Q} &= -\kappa A \frac{dT}{dX} \\
\frac{dT}{dx} &= -\frac{\dot{Q}}{\kappa A} \\
-\int \frac{\dot{Q}}{\kappa A} dx &= \int dT \\
-\int_0^L \frac{\dot{Q}}{\kappa A} &= \int_{T_1}^{T_2} dT \\
-\frac{\dot{Q}}{\kappa A} [x]_0^L &= T_2 - T_1 \\
\frac{dT}{dx} &= \frac{T_2 - T_1}{L}
\end{aligned}$$

2.1 Thermal Conductivity

A higher value of κ means a material is more thermally conductive, for example:

- Diamond (natural): $\kappa = 2200\text{Wm}^{-1}\text{K}^{-1}$.
- Copper: $\kappa = 400\text{Wm}^{-1}\text{K}^{-1}$.
- Air: $\kappa = 0.02\text{Wm}^{-1}\text{K}^{-1}$.
- Aerogel: $\kappa = 0.003\text{Wm}^{-1}\text{K}^{-1}$.

Diamond is highly thermally conductive due to high frequency vibrations ("phonons") being able to propagate through the material as a result of its rigidity.

There are two sources of heat conductivity, one being as a result of electrons and one as a result of phonons.

- Inside the material, we have two bands - valence and conduction. These bands are a thick collection of many energy levels very close together. As the Pauli exclusion principle says that no two electrons can share a quantum number, each level must shift infinitesimally to not overlap. The levels within each band are so close that we treat electron as being able to move freely between them.
- The valence band is where electrons are still bound to an atom, the conduction band is where they are bound to the material as a whole and are free to move through the material.
- For an insulator, the difference between the two bands is $\sim$ 10eV. This represents a very small proportion of particles with sufficient energy and results in very little thermal excitation (hence little thermal conduction). As an electron is excited, it leaves a hole behind (that we treat as a particle h^+ for ease). We treat this hole as being able to move (as electrons themselves move).

[1] We also have another very similar quantity labelled K which will be covered later and may be a source of confusion

- For an electron to fall back to the valence band, it must fall into a hole. In an insulator the conduction band is effectively empty. For a semiconductor, this gap is more like $\sim$ 1eV, so a much higher number of electrons can jump, but this is still temperature dependant.
- In metals, the conduction and valence bands have little to no gap. Therefore, all electrons are free conduction electrons.

Diamonds sit on the boundary between being an insulator and a semiconductor. They work by phonons. As we heat a material, bonds in the material expand and cause a vibration. As diamond is very rigid, vibrating one bond causes a ripple effect - effectively a wave of excitation (where one of these waves is called a phonon). This phonon propagation causes the transfer of vibrations hence the transfer of heat energy.

2.2 Thermal Properties

We can also define the:

- Thermal Resistivity: $\rho = 1/\kappa$.
- Thermal Conductance $K = \kappa A/L$. [2]
- Thermal Resistance: $R = L/(\kappa A)$.

This are the heat analogues of the equivalent terms for electrical conductance in a circuit.

Annoyingly, thermal conductivity also varies with temperature...It also does so in a really weird way depending on the material:

[2] This is the aforementioned annoying symbol overlap between κ and K

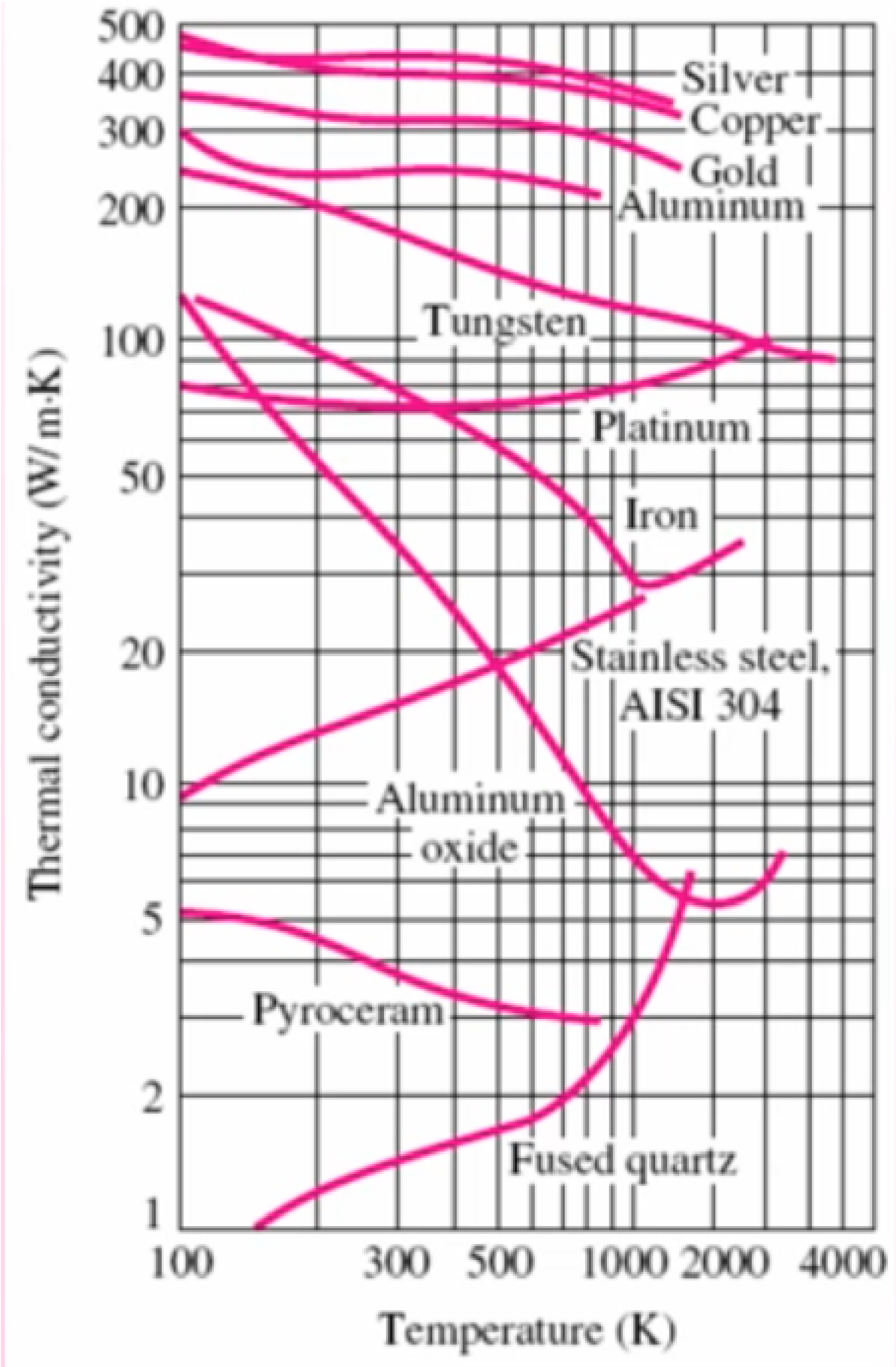

Figure 12.2: κ for various materials, as a function of temperature.

This comes from phonon excitation and electron excitation being two different competing process that both have different degrees of contribution at different temperatures.

This means that Newton's Law of Cooling no longer holds for $\kappa = f(T)$, as the proportionality no longer holds...

Tue 03 Mar 2026 12:00

Lecture 13 - Thermal Physics VII: Convection and Black Bodies

Recap from last lecture

- Three methods of heat transport: conduction, convection and radiation.
- Fourier's Law:
$$\dot{Q} = -\kappa A \frac{dT}{dx}$$
- We can show that:
$$\frac{dT}{dx} = \frac{T_2 - T_1}{L}$$
- Conduction determined by two distinct physical processes that transfer heat:
  - Movement of electrons.
  - Photon transfer.
- Thermal resistivity, thermal conductance and thermal resistance, acting as analogues of their electrical equivalents.

1 Conduction II

1.1 Combinations of Thermal Resistances

A derivation of why isn't examinable (but is shown in the non-assessed problems), but we have the same rules for Thermal Resistances as we do in electrical resistances:

- Thermal resistances add in series, $R_T = R_1 + R_2 + \cdots R_n$.
- Thermal resistances add in inverse when in parallel $1/R_T = 1/R_1 + 1/R_2 + \cdots + 1/R_n$.

1.2 Example: Pipe Lagging

Consider a solid copper pipe of radius r_0, initial temperature T_0 and length L. We surround this pipe with insulation of initial temperature T_1 and resulting in a new radius r_1. The insulation has $\kappa = 0.05\text{Wm}^{-1}\text{K}^{-1}$ and the system is in a steady state.

We start from Fourier's law:
$$\dot{Q} = -\kappa A \frac{dT}{dr}$$

As the pipe is a cylinder:
$$\dot{Q} = -\kappa(2\pi r l)\frac{dT}{dr}$$
$$-\int_{r_0}^{r_1} \frac{\dot{Q}}{2\kappa\pi r l} = -\int_{T_0}^{T_1} dT$$
$$\int_{r_1}^{r_0} \frac{\dot{Q}}{2\kappa\pi l}\frac{dr}{r} = \int_{T_1}^{T_0} dT$$

$$\frac{\dot{Q}}{2\kappa\pi l}[\ln r]_{r_0}^{r_1} = [T]_{T_1}^{T_0}$$

$$\frac{\dot{Q}}{2\kappa\pi l}\ln\left(\frac{r_0}{r_1}\right) = T_0 - T_1$$

$$\dot{Q} = (T_0 - T_1)\frac{2\kappa\pi l}{\ln(r_1/r_0)}$$

2 Convection

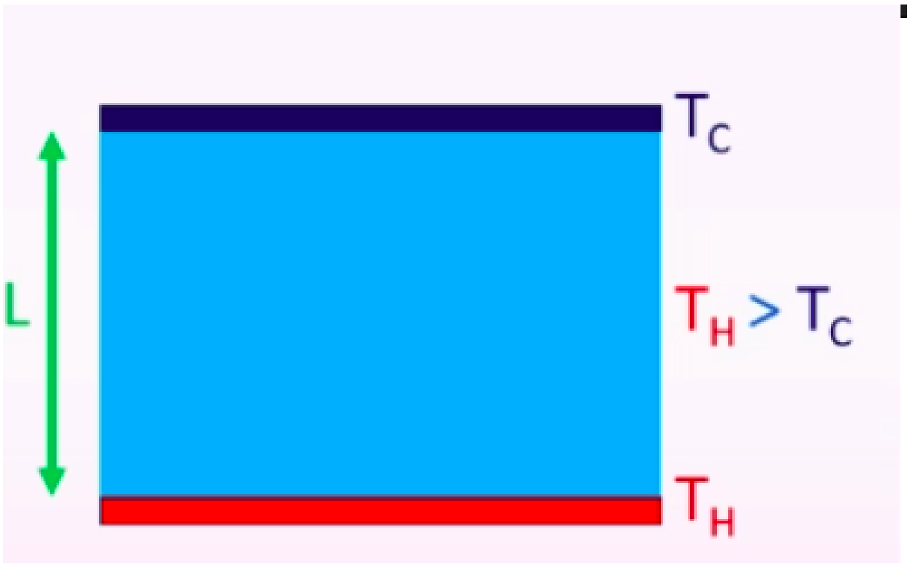

Figure 13.1

Consider a fluid suspended between two plates, distance L apart. The upper plate T_C is colder than the lower plate, T_H.

What governs whether or not convection can occur (and to what extent)?

- Buoyancy of the fluid $\beta = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P$
- The thermal conductivity, κ.
- The viscosity of the fluid, η.

To decide whether or not convection can occur, we use the Rayleigh number:

$$R_A = \frac{\rho L^3(\beta\{T_H - T_C\})g}{\eta\alpha}$$

Where:

$$\alpha = \frac{\kappa}{\rho C_P}$$

If $R_a > 1,000$ then convection occurs laminarly, and if $R_a > 100,000$ then convection occurs turbulently convection does not occur $R_A < 1,000$. Laminar flow is simple, predictable and consistent flow (i.e. straight out of a well made tap). Turbulent flow is much more complex and is pseudorandom. We can predict statistical effects, but turbulent flow is too complex to predict accurately to a very low level.

The details of convection (except for the basic concept) are not examinable.

3 Thermal Radiation

All bodies emit EM radiation characteristic of the object temperature (and not the temperature of the surroundings). The frequency is temperature dependant. We often use infrared (i.e. thermal imaging) as the frequencies are emitted by an object around the temperature of a human body are IR.

A black body perfectly emits and absorbs radiation of all wavelengths, without any reflection. A while body, on the other hand, perfectly reflects all incident light. A black body in thermal equilibrium with surroundings emits exactly the same power spectrum as it absorbs.

Classical mechanics predicts the Rayleigh-Jeans law for emission:

$$u(\lambda, T) = \frac{8\pi k_B T}{\lambda^4}$$

Where u is power emitted per unit area, T is temperature and λ is the wavelength of emitted light. As this is inversely proportional to lambda, this causes the UV catastrophe, where $u \to \infty$ as $\lambda \to 0$. Max Plank developed an improved version[1].

$$u(\lambda, T) = \frac{8\pi hc}{\lambda^5} \frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1}$$

We can characterise non-black bodies by the emissivity, ϵ, the fraction of the black body spectrum which it absorbs. This gives $\epsilon = 1$ for a true black body and $\epsilon = 0$ for a true white body.

4 Wein's Approximation

We want to determine the maximum of the spectrum, to find the wavelength corresponding to the maximal power output.

$$u(\lambda, T) = \frac{8\pi hc}{\lambda^5} \frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1}$$

Letting $a = \frac{hc}{\lambda k_B T}$:

$$\begin{aligned} \frac{1}{\lambda} &= \frac{k_B T}{hc} a \\ \implies \frac{1}{\lambda^5} &= \left(\frac{k_B T}{hc}\right)^5 a^5 \\ \implies \frac{8\pi hc}{\lambda^5} &= 8\pi hc \left(\frac{k_B T}{hc}\right)^5 a^5 \end{aligned}$$

And:

$$\frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1} = \frac{1}{\exp(a) - 1}$$

$$\frac{\partial a}{\partial \lambda} = \frac{-hc}{\lambda^2 k_B T} = -\frac{1}{\lambda} \frac{hc}{\lambda k_B T} = -\frac{a}{\lambda}$$

Hence:

$$u(\lambda, T) = \underbrace{8\pi hc \left(\frac{k_B T}{hc}\right)^5}_{\text{constant } = C} a^5 \times \frac{1}{\exp(a) - 1}$$

$$u(\lambda, T) = \frac{Ca^5}{e^a - 1}$$

$$\begin{aligned} \frac{\partial u}{\partial \lambda} &= \frac{\partial a}{\partial \lambda} \frac{d}{da} \left[\frac{Ca^5}{e^a - 1} \right] \\ &= \frac{-a}{\lambda} \frac{d}{da} \left[\frac{Ca^5}{e^a - 1} \right] \end{aligned}$$

[1] See QM1

Using the quotient rule:

$$\frac{d}{da}\left[\frac{Ca^5}{e^a-1}\right] = \frac{5Ca^4(e^a-1) - Ca^5e^a}{(e^a-1)^2}$$

Therefore:

$$\frac{\partial u}{\partial \lambda} = \frac{-a}{\lambda}\left[\frac{5Ca^4(e^a-1) - Ca^5e^a}{(e^a-1)^2}\right]$$

Solving for the maxima:

$$\frac{\partial u}{\partial \lambda} = 0$$

$$-a\left(5Ca^4(e^a-1) - Ca^5e^a\right) = 0$$

As C is a non-zero constant:

$$-a\left(5a^4(e^a-1) - a^5e^a\right) = 0$$

As $a \neq 0$:

$$5a^4(e^a-1) - a^5e^a = 0$$

Dividing by a^4:

$$5(e^a-1) - ae^a = 0$$

$$5e^a - 5 - ae^a = 0$$
$$(5-a)e^a = 5$$

Solving numerically:

$$a \approx 4.965$$
$$a = \frac{hc}{\lambda k_B T}$$
$$\lambda_{\max} T = \frac{hc}{k_B a}$$

Finally:

$$\lambda_{\max} T \approx 2.898 \times 10^{-3}\,\mathrm{m\,K}$$

Thu 05 Mar 2026 13:00

Lecture 14 - Statistical Physics I: Introduction

1 Stefan-Boltzmann Law

We can integrate across the whole spectrum to work out total energy emitted per unit time:

$$\dot{Q} = \epsilon\sigma A T^4$$

Where $\sigma = 5.67 \times 10^{-8}\text{Wm}^{-2}\text{K}^{-1}$ is the Stefan-Boltzmann constant, ϵ is the emissivity, A is the surface area of the radiating body and T is the temperature.

2 Kinetic Theory of Gases

We've now finished the module on temperature and move into statistical physics. We start wth a kinetic theory of the motion of gas molecules. Gas molecules move "randomly" (effectively, it's not computationally viable to determine what's actually going on), with vastly different velocities and positions.

We would love to describe a gas using the individual molecules that make it up, but that's really difficult to do for $\sim 10^{23}$ molecules so it's not viable and we have to use statistical physics to describe it.

We've previously touched upon the Maxwell-Boltzmann distribution to describe the velocities of particles as a statistical distribution wrt temperature:

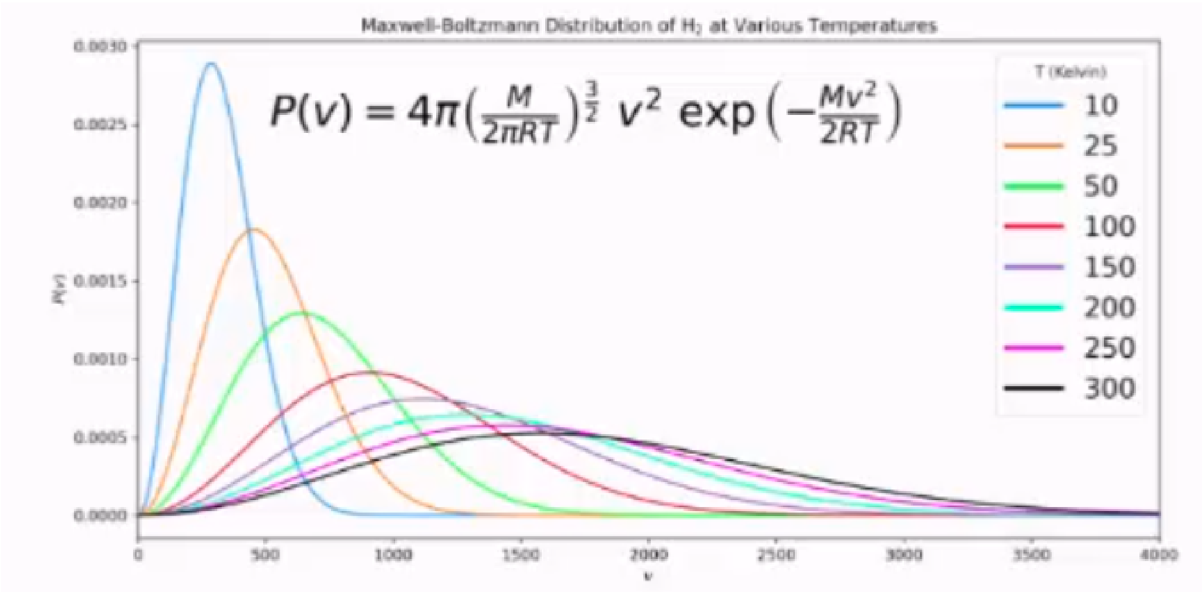

Figure 14.1

These statistical approximations improve as we increase the number of particles, and for very large numbers they become good approximations. We can in statistical physics have a probability > 1. This seems mathematically strange, but we interpret it to mean that (if the probability represents the probability of an event occurring in a measurement), the event will take place multiple times. We still cannot have any probabilities < 0.

The rest of the lecture then turned into a fun tangent on statistics which isn't examinable and needs no notes.

Tue 10 Mar 2026 12:00

Lecture 15 - Statistical Physics II: The Atmosphere

1 The Atmosphere

The very outer part of the atmosphere, from 90km to 350km is the ionosphere. This acts as a mirror which reflects EM radiation

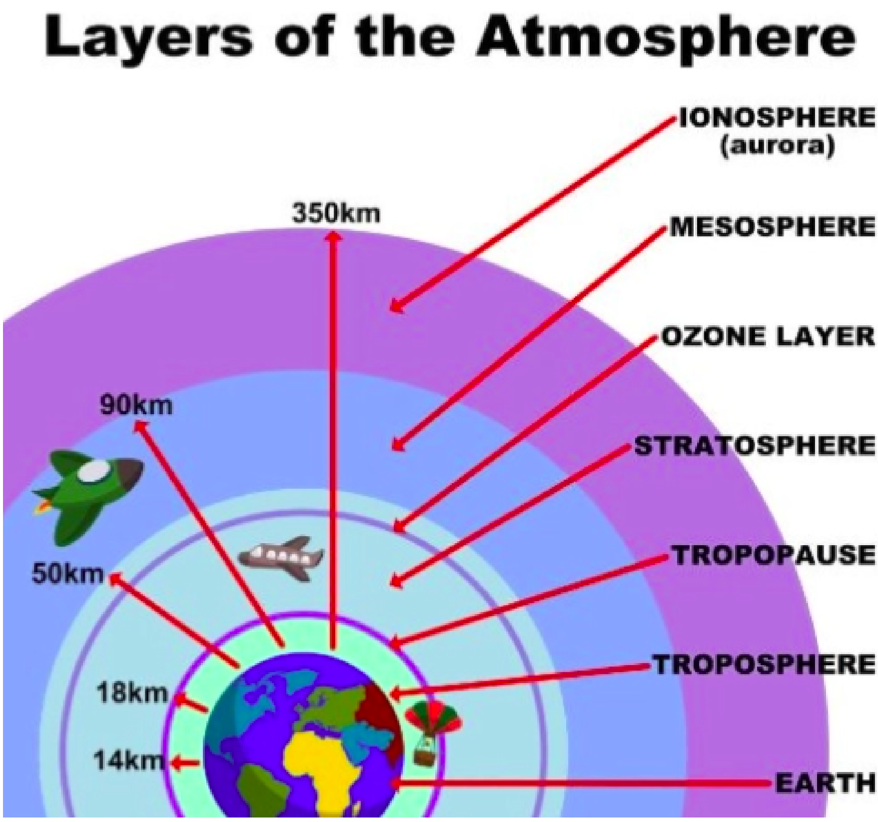

Figure 15.1

This is how we're able to do international radio communications, by bouncing radio waves off of this layer in order to reach continents where there is no direct line of sight.

1.1 Isothermal Model of the Atmosphere

Consider an observer sitting in the troposphere above the earth, looking at a small cuboidal slice of the troposphere.

We will assume that:

- The atmosphere is isothermal. In reality, increasing height reduces temperature by approx. 10K per km.
- The atmosphere is solely comprised of an ideal gas. At the low pressure scales relevant for the atmosphere, this is a reasonable assumption.

- The earth is flat[1]. Technically, assume that we we consider small enough heights and cross-sectional areas that we can disregard the curvature of the earth.
- The atmosphere is stationary and is therefore in mechanical equilibrium. This can be a good approximation depending on weather conditions.

The slab of atmosphere we're considering has area A, height dh and sits at height h above the surface of the earth. From the ideal gas state equation:

$$PV = \underbrace{nRT}_{\text{moles}} = \underbrace{Nk_BT}_{\text{particles}}$$

$$\implies P = \frac{N}{V}k_BT$$

$$\implies P = \rho_N k_B T$$

Where $\rho_N = N/V$ is number density, and is defined as the number of molecules per unit volume.

The slab of atmosphere has two forces acting upon it, a force due to pressure and a force due to gravity. As we assume that slice is in mechanical equilibrium, we can equate these forces.

The pressure is going to vary across the slab. Force due to pressure increases the further down into the atmosphere you go. This means the bottom face of the slab experiences a higher pressure than the top face. Since the slab is in mechanical equilibrium, the force due to gravity must balance the net upwards pressure force.

This gives us:

$$\text{force due to gravity} = \text{force due to pressure}$$

$$mg\rho_N V = AP(h) - AP(h+dh)$$

$$mg\rho_N V = -dP(h)A$$

$$mg\rho_N \times dh \times A = -d\rho(h)A$$

$$mg\rho_N \times dh = -d\rho(h)$$

$$\frac{d\rho(h)}{dh} - mg\rho(N)$$

$$\int d\rho(h) = -\int mg\rho_N dh$$

And considering number density in the density integral:

$$\int_{\rho(0)}^{\rho(h)} \frac{1}{\rho_N} d\rho(h) = -\int_0^h mgdh$$

And substituting $dp = d\rho_N k_B T$ to give a LHS integral entirely in number density:

$$\int_{\rho_N(0)}^{\rho_N(h)} \frac{k_BT}{\rho_N} d\rho_N = -mgh$$

$$\int_{\rho_N(0)}^{\rho_N(h)} \frac{k_BT}{\rho_N} d\rho_N = -\frac{mgh}{k_BT}$$

$$[\ln(\rho_N)]_{\rho_N(0)}^{\rho_N(h)} = -\frac{mgh}{k_BT}$$

$$\frac{\rho_N(h)}{\rho_N(0)} = \exp\left(-\frac{mgh}{k_BT}\right) \implies \rho_N(h) = \rho_N(0)\exp\left(-\frac{mgh}{k_BT}\right)$$

The contents of the exponential have to be unitless, and they are because they're both units of energy. The upper gives gravitational energy, the lower gives thermal energy and the contents of the exponential depends on the ratio between the two of them.

[1] Oh dear...

This gives the final result:

$$\boxed{\rho_N(h) = \rho_N(0)\exp\left(-\frac{mgh}{k_BT}\right)}$$

And as $P = \rho_n k_B T$:

$$P(h)k_BT = P(0)k_BT\exp\left(-\frac{mgh}{k_BT}\right)$$

$$\implies \boxed{P(h) = P(0)\exp\left(-\frac{mgh}{k_BT}\right)}$$

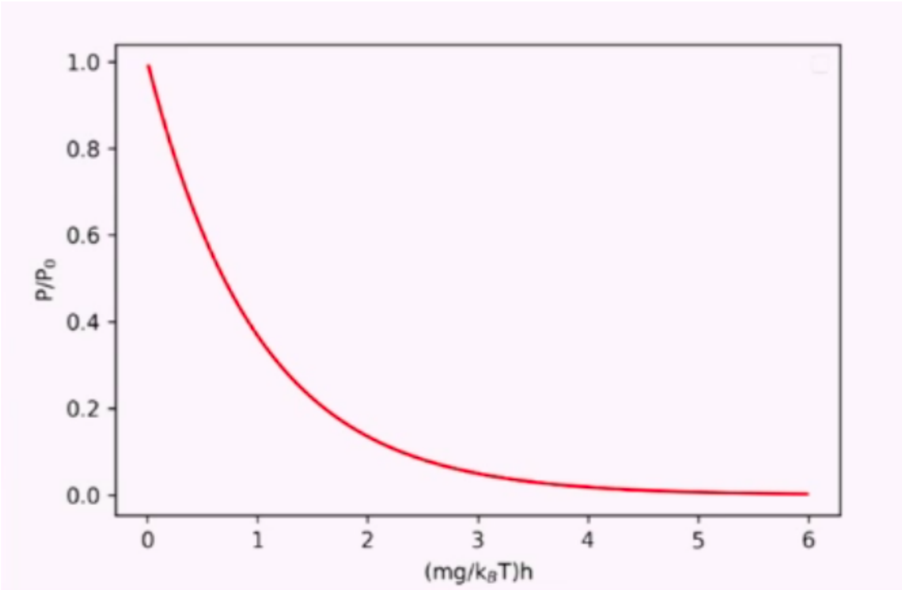

Figure 15.2

1.2 Scale Height

As mgh/k_BT must be dimensionless, mg/k_BT must have units of h. We call this the scale height, h_0. This is the height at which $P(h_0) = P(0)e^{-1}$, i.e. the point at which pressure drops off by a factor of $1/e$.

Particle	Mass	Scale height
O_2 gas	32 amu = 5.3137e-26 kg	~10 km
N_2 gas	28 amu = 4.6495e-26 kg	~10 km
Covid virus	~10^{-18} kg	~0.5 mm
Smallest known virus	~10^{-21} kg	~50 cm
Bacteria	~10^{-15} kg	~5 μm
Dr Stuart Pirrie	~10^{2} kg	~50 ym (10^{-24} m is 1 ym)

Figure 15.3: Example Scale Heights

This helps answer the question "if someone coughs on a desk, are other people in the room safe from Covid?". This implies yes, as the scale heigh is much larger than anyone is likely to be above a desk. This isn't true for all viruses however, i.e. the 50cm scale height for the smallest known virus.

Thu 12 Mar 2026 13:00

Lecture 16 - Statistical Physics III: Atmosphere and Boltzmann Factors

Recap from last lecture:

- We derived:

$$\rho_N(h) = \rho_N(0) \exp\left(-\frac{mgh}{k_B T}\right)$$
$$P(h) = P(0) \exp\left(-\frac{mgh}{k_B T}\right)$$

  To give the number density and pressure as an exponential function of height in the atmosphere.

- Since $mgh/k_B T$ is being passed into an exponential, it must be dimensionless. Hence $k_B T/mg$ must have dimensions of h.
- This leads to the scale height h_0 where $P(h_0) = P(0)/e$

1 Atmosphere Continued

We have established that the number density (number of fluid molecules per unit volume), at height 0, $\rho_N(0)$ can be related to atmospheric pressure ($P(0)$) by:

$$\rho_N(0) = \frac{P(0)}{k_B T}$$

To determine the total number of gas molecules in the slab of atmosphere with cross-sectional area A, we can integrate number density across all h:

$$\begin{aligned}
N &= A \int_0^\infty \rho_N(h)\, dh \\
&= A \int_0^\infty \rho_N(0) \exp\left(-\frac{mgh}{k_B T}\right) dh \\
&= A\rho_N(0) \int_0^\infty \exp\left(-\frac{mgh}{k_B T}\right) dh \\
&= A\rho_N(0) \left[-\frac{k_B T}{mg} \exp\left(-\frac{mhg}{k_B T}\right)\right]_0^\infty \\
&= -A\rho_N(0) \frac{k_B T}{mg} [0 - 1] \\
N &= A\rho_N(0) \frac{k_B T}{mg} \\
\implies \rho_N(0) &= \frac{Nmg}{Ak_B T}
\end{aligned}$$

And using:

$$h_0 = \frac{k_B T}{mg}$$
$$\rho_N(0) = \frac{N}{Ah_0}$$

This has a physical interpretation. If you took all N molecules and distributed them in a volume Ah_0, i.e. our slab of cross-sectional area A with $h = h_0$, the average density of this would be equal to $\rho_N(0)$. This defines h_0 additionally as the average height of an equivalent uniform atmosphere with the same number of molecules.

We further define a new quantity:

$$\Pr(h) = \frac{A\rho_N(h)}{N}$$

Which gives the probability of (for a given specific particle) finding it at height h

1.1 Probability and Normalisation

We start from:

$$\Pr(h) = \frac{A\rho_N(h)}{N}$$

$$h_0 = \frac{k_B T}{mg}$$

$$\rho_N(h) = \rho_N(0)\exp(-mgh/k_B T)$$

And substitute our previous results for $\rho_N(0)$ and $\rho_N(h)$ in:

$$\Pr(h) = \frac{A}{N} \times \rho_N(0)\exp(-mgh/k_B T)$$

$$\Pr(h) = \frac{A}{N} \times \frac{N}{Ah_0}\exp(-mhg/k_B T)$$

$$\Pr(h) = \frac{mg}{k_B T}\exp(-mhg/k_B T)$$

As this is a probability, we need to check that it is normalised:

$$\int_0^\infty \Pr(h)\,dh$$

$$= \int_0^\infty \frac{A\rho_N(h)}{N}\,dh$$

$$\frac{A}{N}\int_0^\infty \rho_N(h)\,dh$$

$$\int_0^\infty \Pr(h)\,dh = \frac{1}{N}\left(A\int_0^\infty \rho_N(h)\,dh\right)$$

And as previously derived:

$$A\int_0^\infty \rho_N(h)\,dh = N$$

So:

$$\int_0^\infty \Pr(h)\,dh = \frac{1}{N}\left(A\int_0^\infty \rho_N(h)\,dh\right) = \frac{N}{N} = 1$$

So $\Pr(h)$ is normalised, as required!

2 Boltzmann Factors

This is on the exam. We have the quantity mgh/k_BT, which is dimensionless. What does this quantity actually physically mean? mgh is a term for the potential energy (gravitational here), and k_BT is a term for thermal energy.

Boltzmann factors describe the probabilities of measuring a given energy state in a system. In general form:

$$\Pr(E_i) \propto \exp\left(-\frac{E_i}{k_BT}\right)$$

This is true for all energy states, i.e. a mass on a spring oscillating in SHM has:

$$\Pr(\text{state where extension is } x) \propto \exp\left(-\frac{0.5kx^2}{k_BT}\right)$$

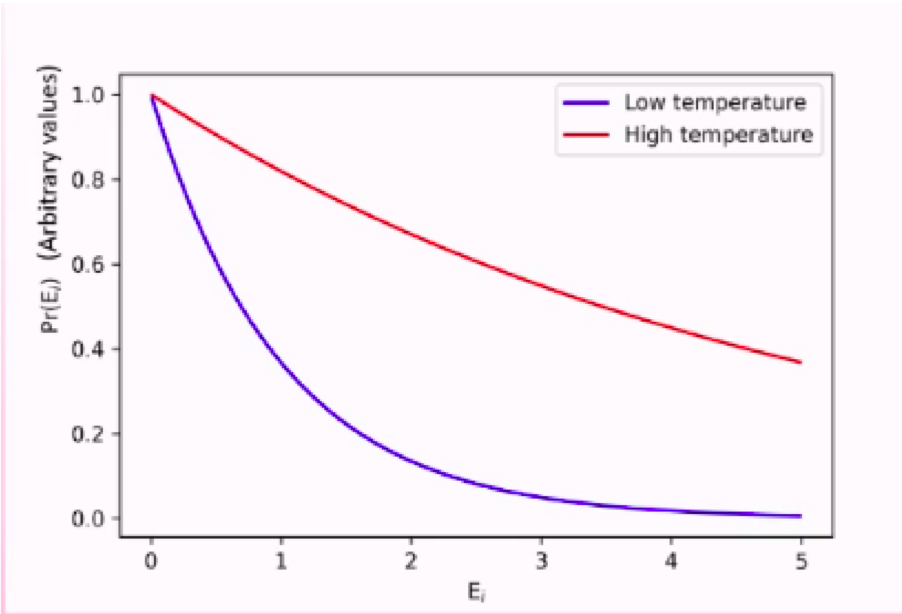

Figure 16.1

From this figure, we can see that higher temperatures have a higher probability of observing higher energy states, which makes sense. We can use this to describe continuous energy distributions, or discrete ones (i.e. energy levels in atoms).

The integral across all energy states must be equal to one.

3 Example: Simple Quantum System

Consider a simple atom with two energy levels E_0 and E_1.

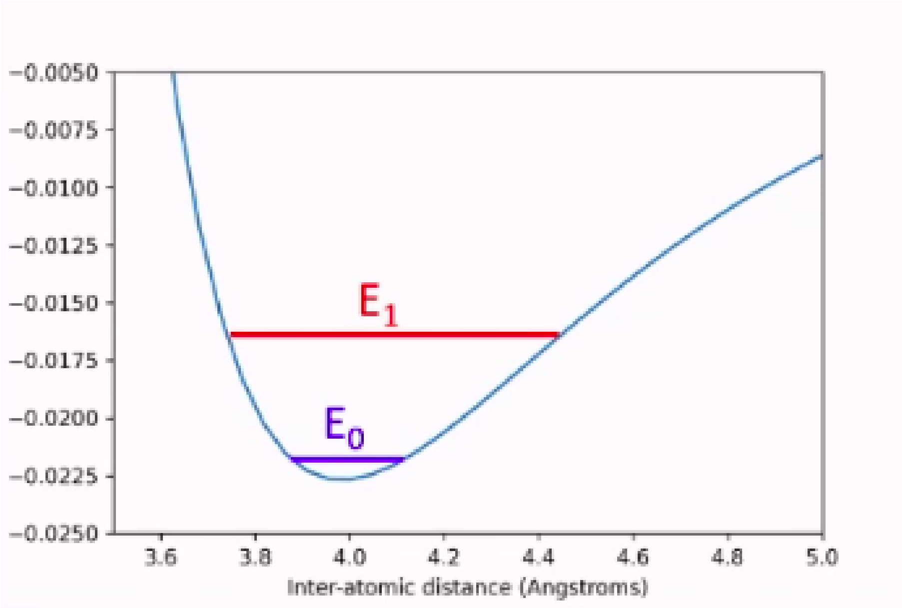

Figure 16.2

What is the probability of finding the atom at energy state E_0?

$$\Pr(E_0) \propto \exp\left(-\frac{E_0}{k_B T}\right)$$

Or adding a constant of proportionality:

$$\Pr(E_0) = C\exp\left(-\frac{E_0}{k_B T}\right)$$

And if both energy states have the same constant of proportionality, which is true here:

$$\Pr(E_1) = C\exp\left(-\frac{E_1}{k_B T}\right)$$

Since the probability must be normalised across all states:

$$\Pr(E_0) + \Pr(E_1) = 1$$

$$C\exp\left(-\frac{E_0}{k_B T}\right) + C\exp\left(-\frac{E_1}{k_B T}\right) = 1$$

$$1 = C\left(\exp\left(-\frac{E_0}{k_B T}\right) + \exp\left(-\frac{E_1}{k_B T}\right)\right)$$

$$C = \frac{1}{\exp(-E_0/k_B T) + \exp(-E_1/k_B T)}$$

Is the constant of proportionality here. Hence:

$$\Pr(E_1) = \frac{\exp(-E_1/k_B T)}{\exp(-E_0/k_B T) + \exp(-E_1/k_B T)}$$

$$\Pr(E_0) = \frac{\exp(-E_0/k_B T)}{\exp(-E_0/k_B T) + \exp(-E_1/k_B T)}$$

We can now do a bit of a maths trick:

$$\Pr(E_1) = \frac{\exp(-E_1/k_B T)}{\exp(-E_0/k_B T) + \exp(-E_1/k_B T)} \times \frac{e^{E_1/k_B T}}{e^{E_1/k_B T}}$$

$$\frac{1}{\exp\left(\frac{E_1-E_0}{k_BT}\right)+1}$$

Which lets us start considering limits. If we consider the limit as $T \to 0$:

$$\Pr(E_1) \to 0 \text{ and } \Pr(E_0) \to 1$$

The maths is unpleasant, but we can take it as given! This implies that in the low temperature limit, the atom always sits in the non-excited state.

What about the high temperature limit? We begin with a Taylor expansion:

$$\Pr(E_1) \approx \frac{1}{1+(E_1-E_0)/k_BT+1}$$

Now as $T \to \infty$:

$$\Pr(E_1) \to \frac{1}{1+0+1} \implies \Pr(E_1) \to \frac{1}{2} \text{ and } \Pr(E_0) \to \frac{1}{2}$$

So in the high temperature limit, we end up with an even distribution between the two states.

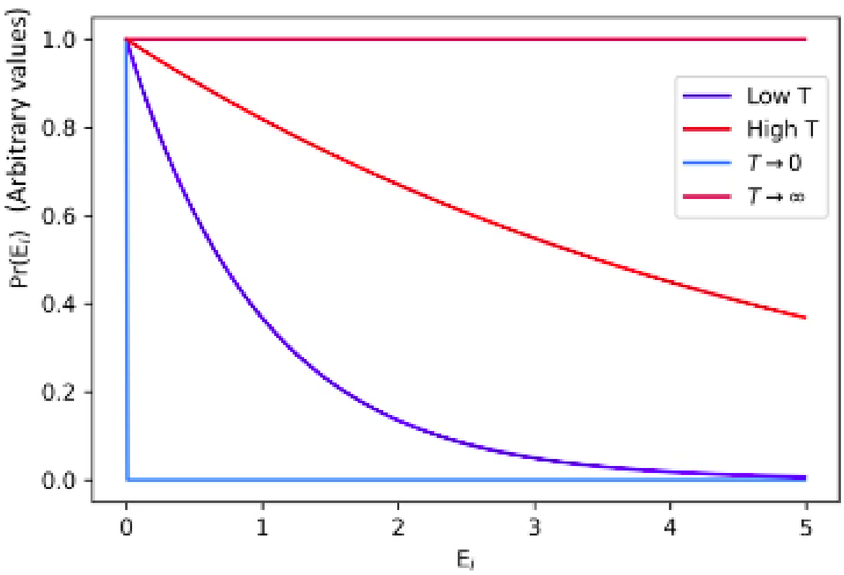

Figure 16.3

4 Negative Temperatures

Tue 17 Mar 2026 12:00

Lecture 17 - Statistical Physics IV: Boltzmann Factors and Degeneracies

Recap from last lecture:

- We can predict the microscopic behaviour of a system using a probability density function defined using macroscopic terms.
- We can predict the likelihood of a given energy state begin populated for a system (discrete or continuous) with the Boltzmann factor.
- For a two level system with $E_1 < E_0$ we have:

$$\Pr\left(E_i\right) = \frac{\exp\left(\frac{-E_i}{k_B T}\right)}{\exp\left(\frac{-E_0}{k_B T}\right) + \exp\left(\frac{-E_1}{k_B T}\right)}$$

- While not explicitly covered last lecture, this trivially generalises to a system with N levels:

$$\Pr\left(E_i\right) = \frac{\exp\left(\frac{-E_i}{k_B T}\right)}{\sum_{j=0}^{N-1} \exp\left(\frac{-E_j}{k_B T}\right)}$$

1 Degeneracies

In an atom, electrons can have the same energy in multiple different combinations (multiple ways of having the same result). For example, considering only one energy level, electrons can be spin up or spin down. This gives two possible options, hence a degeneracy of two.

With electrons in shells, the degeneracy gives the number of electrons which can fit in a shell. The Pauli exclusion principle says that no two atoms can occupy a shell if they share a quantum number, so the number of possible combinations of distinct states is equal to the number of possible electrons in the shell.

Given the degeneracy $g(E)$, we can modify the Boltzmann Factor accordingly:

$$\Pr(E_i) \propto g(E) \exp\left(\frac{-E_i}{k_B T}\right)$$

For $n = 1$, $g(E) = 2$, for $n = 2$, $g(E) = 4$, and so on.

2 Boltzmann Factors and Ideal Gases

For an ideal gas, we have some assumptions (in terms of energy) that have been discussed previously. The most important of these in terms of energy is:

- Internal energy all comes from the average kinetic energy of the molecules (temperature) and not from potential energy.
- The Boltzmann Factor for an ideal gas hence takes the form:

$$\Pr(E_i) \propto \exp\left(\frac{-E_K}{k_B T}\right)$$

Our other assumptions are:

1. The gas is in thermal equilibrium, hence $\langle E_K \rangle$ is constant.
2. No gravity (as molecules are extremely light).
3. Assume that the average separation of each molecule is much larger than the molecule size, hence collisions are infrequent.
4. There is a very large number of molecules $\sim 10^{23}$ to allow us to treat the gas statistically without high statistical uncertainties.
5. All molecules have the same mass and size. This is not a good assumption for gases made up of multiple component gases mixed.
6. We assume that the direction of molecular motion is effectively random, i.e. is unbiased to any particular direction.
7. Molecules can collide, but do so elastically (and as previously mentioned, infrequently).
8. Molecules are not relativistic, so $E_K = 0.5mv^2$ holds.

3 Model of Ideal Gas

We can split the velocity of a gas into three cartesian coordinates by adding in quadrature:

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

By assumption 6, we assume that all of these components are effectively equal, so we can consider the x-direction alone initially (and extrapolate out).

We want to consider the probability of having a velocity somewhere between v_x and $v_x + dx$.

$$\Pr(v_x \to v_x + dx) = \int_{v_x}^{v_x+dx} \Pr(v_x)\, dv_x$$

$$= \int_{v_x}^{v_x+dx} C \exp\left(-\frac{0.5mv_x^2}{k_B T}\right) dv_x$$

And using the standard integral: $\int_{-\infty}^{\infty} e^{-ax^2}\, dx = \sqrt{\pi/a}$, we have:

$$\int_{-\infty}^{\infty} C \exp\left(-\frac{mv_x^2}{2k_B T}\right) dv_x = 1$$

$$\implies C\sqrt{\frac{\pi}{a}} = 1$$

And as $a = m/2k_B T$:

$$C\sqrt{\frac{2\pi k_B T}{m}} = 1$$

$$\implies C = \sqrt{\frac{m}{2\pi k_B T}}$$

And as the Boltzmann factor for this x-directional motion is $\exp\left(mv_x^2/2k_B T\right)$:

$$\Pr(v_x) = C \exp\left(\frac{mv_x^2}{2k_B T}\right)$$

We finally have:

$$\Pr(v_x) = \sqrt{\frac{m}{2\pi k_B T}} \exp\left(-\frac{0.5mv_x^2}{k_B T}\right)$$

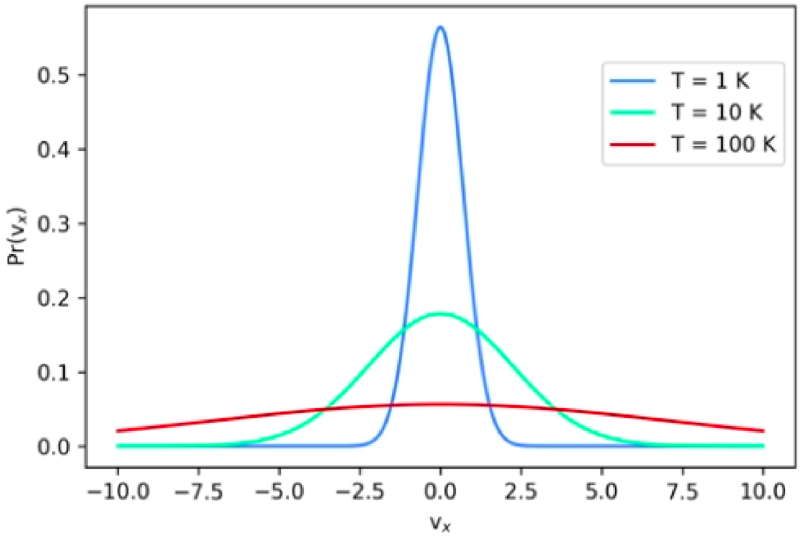

Figure 17.1

As we're experiencing random gas motion, the distribution is centred on 0, as velocity can be positive or negative.

As we increase the temperature, we increase the chance of occupying higher energy states (i.e. higher v_x).

This distribution has no degeneracies, as there's only one way for a particle to have a specific v_x, which is by travelling at that value. It either has a specified velocity, or it simply doesn't.

3.1 3D Model

Now we have a single component:

$$\Pr(v_x) = \sqrt{\frac{m}{2\pi k_B T}} \exp\left(-\frac{0.5 m v_x^2}{k_B T}\right)$$

We can extend this to 3D:

$$\Pr(v) = \Pr(v_x)\Pr(v_y)\Pr(v_z)$$

$$\Pr(v) = \left(\frac{m}{2\pi k_B T}\right)^{(3/2)} \exp\left(-\frac{0.5m(v_x^2 + v_y^2 + v_z^2)}{k_B T}\right)$$

3.2 Density of States

While we had no degeneracies in each component direction, we do have degeneracies for v itself. For example, $v^2 = 100$ could mean that x, y are zero and $z = 10$, or any other number of combinations.

We therefore want to define the degeneracies as a function of v. As a model, consider a grid of infinitesimally small dots in 2D (but for the actual application we would want a 3D model).

Our final velocity in 2D is $v^2 = v_x^2 + v_y^2$, so if each axis represents possible values for v_x and v_y, v is given as the circumference of a circle with radius v. The density of states $g(v)dv$ is given as the ring formed between v and $v + dv$.

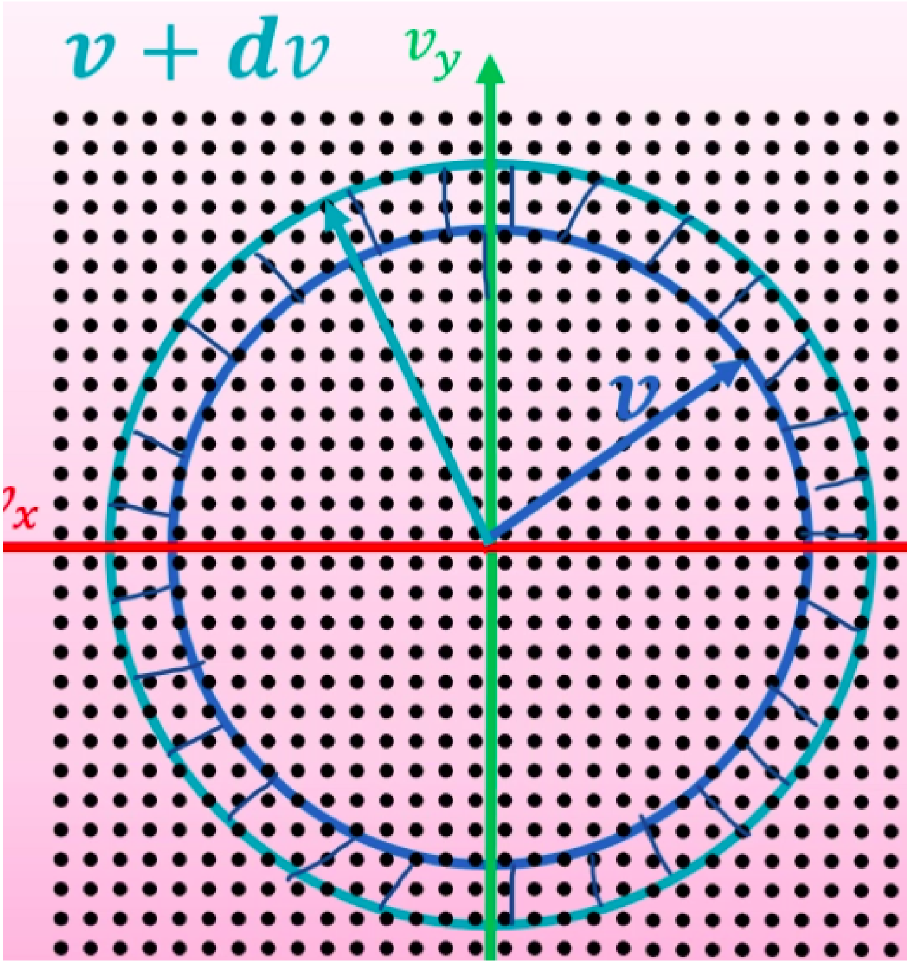

Figure 17.2

In 2D, the degeneracy is therefore given by:

$$g(v)dv = 2\pi v dv$$

In 3D, rather than a ring this would be a spherical shell surface:

$$g(v)dv = 4\pi v^2 dv$$

Applying this to the Maxwell-Boltzmann distribution, we obtain:

$$\Pr(v) = 4\pi \left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 \exp\left(-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right) \qquad \text{where: } v \neq 0$$

And as $v^2 = v_x^2 + v_y^2 + v_z^2$:

$$\Pr(v) = 4\pi \left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2k_B T}\right) \qquad \text{where: } v \neq 0$$

This finally gives us the Maxwell-Boltzmann distribution we encountered before (although we have derived it for molecules vs moles):

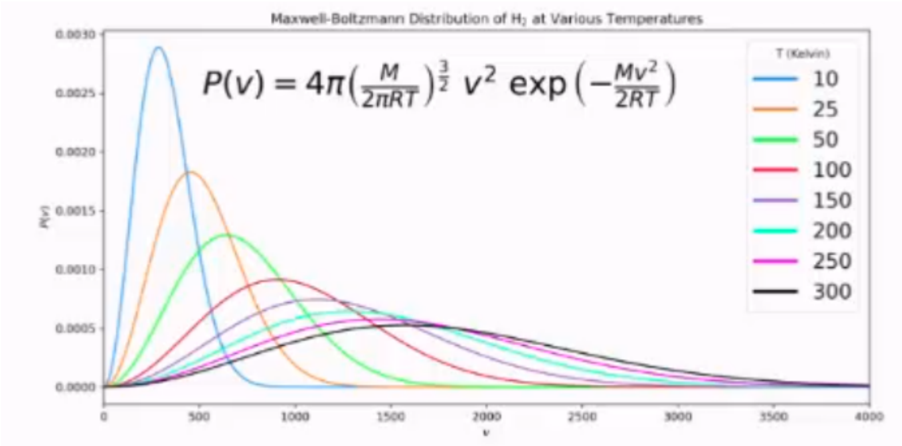
Maxwell-Boltzmann Distribution of H_2 at Various Temperatures
$P(v) = 4\pi\left(\frac{M}{2\pi RT}\right)^{\frac{3}{2}} v^2 \exp\left(-\frac{Mv^2}{2RT}\right)$
T (Kelvin)
10
25
50
100
150
200
250
300
P(v)
v

Figure 17.3

Thu 19 Mar 2026 13:00

Lecture 18 - Statistical Physics V: Maxwell-Boltzmann Distribution

Recap from last lecture:

- We derived:

$$\Pr(v) = g(v)\left(\sqrt{\frac{m}{2\pi k_B T}}\right)^{(3/2)} \exp\left(-\frac{mv^2}{2k_b T}\right)$$

- Where the degeneracy is given by the density of states:

$$g(v) = 4\pi v^2$$

- Giving the full Maxwell-Boltzmann distribution:

1 Using Maxwell-Boltzmann

What is the probability of finding a particle with speed in excess of 400m/s?

$$\Pr(v > 400) = \int_{400}^{\infty} \Pr(v)\, dv$$

Using this distribution, what is the probability of finding a particle in excess of the speed of light?

$$\Pr(v > c) = \int_{c}^{\infty} \Pr(v)\, dv = \text{non zero}$$

Which is non-physical. Luckily, by the time we get close to c, the exponential term becomes so so small that the probability is effectively infinitesimal.

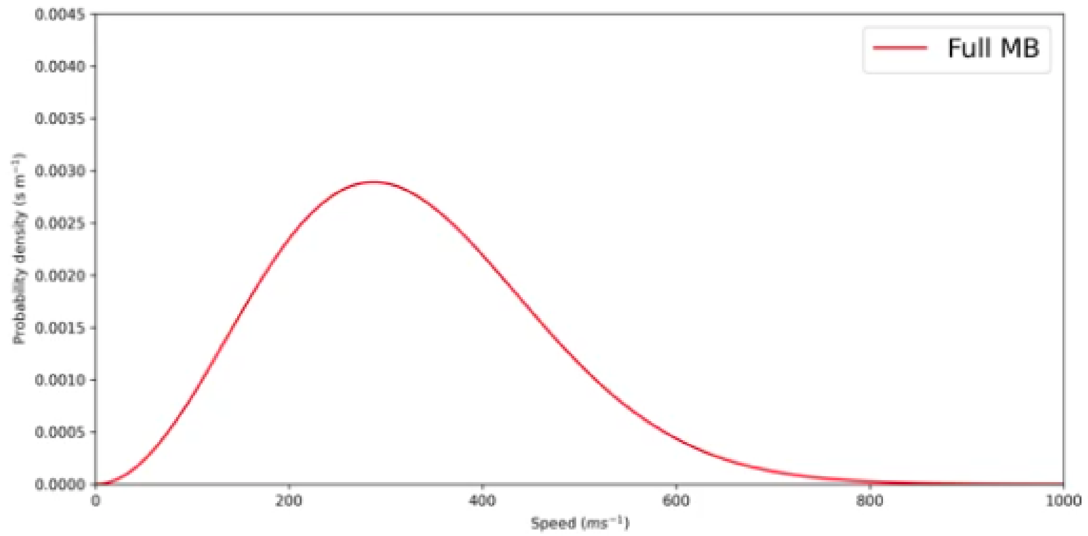

Figure 18.1: Maxwell-Boltzmann Distribution for $T = 10$K. Note that the value is approaching zero for $v \approx 1000$, let alone c.

As the temperature increases, the peak decreases in amplitude and shifts towards higher energy states:

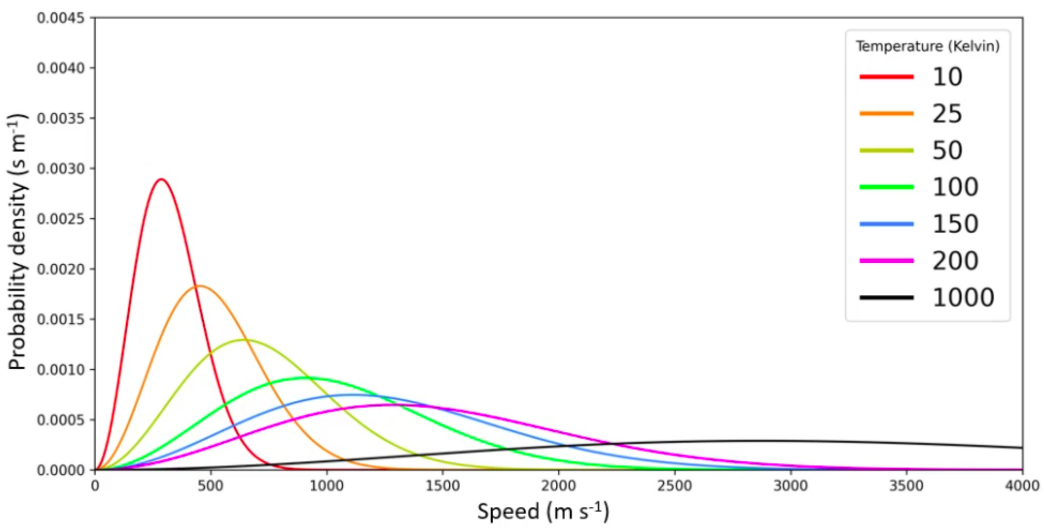

Figure 18.2

Effectively, the probability of occupying higher energy states increases with increasing temperature. What about changing tem

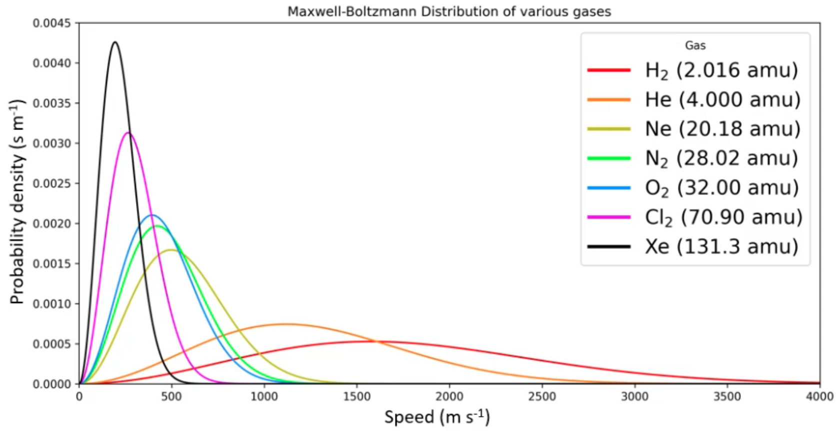

Figure 18.3

This has the opposite effect, which comes from the ratio in the Boltzmann factor.

2 Most Probable Speed

There are two very distinct components to the MBD, when examined componentwise:

$$\Pr(v) = \underbrace{g(v)\left(\sqrt{\frac{m}{2\pi k_B T}}\right)^{(3/2)} v^2}_{\text{quadratic term}} \underbrace{\exp\left(-\frac{mv^2}{2k_b T}\right)}_{\text{exponential term}}$$

This lets us look at the limiting cases:

- As $v \to \infty$:

$$v^2 \to \infty$$
$$\exp\left(-\frac{mv^2}{2k_B R}\right) \to 0$$

  Hence $\Pr(\infty) \to 0$. Showing that the product of an infinite limit and the zero limit would require L'Hopital, but effectively the exponential decays faster than the quadratic grows and wins out.

- As $v \to 0$:

$$v^2 \to \infty \to 0$$
$$\exp\left(-\frac{mv^2}{2k_B T}\right) \to 1$$

  Hence $\Pr(0) \to 0$

As we have $\Pr(0) \to 0$ and $\Pr(\infty) \to 0$, this gives us a curve with a central peak, representing the most probable speed. This speed is give by:

$$\frac{d\Pr(v)}{dv} = 0$$

$$\Pr(v) = 4\pi\left(\sqrt{\frac{m}{2\pi k_B T}}\right)^{(3/2)} v^2 \exp\left(-\frac{mv^2}{2k_b T}\right)$$

$$\Pr(v) = \left[4\pi\left(\sqrt{\frac{m}{2\pi k_B T}}\right)^{(3/2)}\right]\left\{v^2 \exp\left(-\frac{mv^2}{2k_b T}\right)\right\}$$

And performing the product rule on the second {} terms gives:

$$\frac{d\,\mathrm{Pr}(v)}{dv} = 4\pi\left(\sqrt{\frac{m}{2\pi k_B T}}\right)^{(3/2)}\left[2v\exp\left(-\frac{mv^2}{2k_b T}\right)\right] + 4\pi\left(\sqrt{\frac{m}{2\pi k_B T}}\right)^{(3/2)}\left[v^2 \times \frac{-2mv}{2k_B T}\exp\left(-\frac{mv^2}{2k_b T}\right)\right] = 0$$

$$\implies \left[4\pi\frac{m}{2\pi k_B T}^{3/2} 2v\exp\left(-\frac{mv^2}{2k_B T}\right)\right]\left[1 - \frac{mv^2}{2k_B T}\right]$$

The first term can only equal zero in the trivial cases where $v = 0$ or $v = \infty$. If we only consider non-trivial cases, we're left with:

$$1 - \frac{mv^2}{2k_B T} = 0 \implies v = \sqrt{\frac{2k_B T}{m}}$$

For hydrogen gas at 10K, this gives us 287m/s. Does this mean we need to consider supersonic effects (sonic booms) for hotter scenarios?

Luckily, no. With random motion in a gas, there's no obvious direction for pressure to build up in. The effects of gas molecules travelling in lots od different directions cancels itself out nicely.

3 Averages

The average value of x can be written as:

- $\langle x \rangle$.
- $\bar{x}$.
- $E\{x\}$ or $\mathbb{E}\{x\}$.

We'll use the first notation, and strictly it's given by:

$$\langle x \rangle = \frac{1}{N}\sum_{i=1}^{N} N_i x_i = \sum_{i=1}^{N} \Pr_i(x_i) x_i$$

If we consider a histogram, i.e. measurements of velocities for particles. If we take the number of bins to infinity, we have:

$$\langle f(x) \rangle = \int_0^\infty \mathrm{Pr}(x) f(x)\, dx$$

For average **velocity** we expect a distribution which is symmetric about 0, so we expect the average velocity to be 0.

Speed however is given by the Maxwell-Boltzmann distribution and is not symmetric.

$$\mathrm{Pr}(v) = 4\pi\left(\sqrt{\frac{m}{2\pi k_B T}}\right)^{(3/2)} v^2 \exp\left(-\frac{mv^2}{2k_b T}\right)$$

$$\langle v \rangle = \int_0^\infty v\,\mathrm{Pr}(v)\, dv$$

Letting $a = 4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2}$ and $b = \frac{m}{k_B T}$ gives:

$$\langle v \rangle = a\int_0^\infty v^3 e^{-bv^2}\, dv$$

And making the substitution $u = v^2 \implies du = 2vdv$:

$$\langle v \rangle = \frac{a}{2}\int_0^\infty u e^{-bu}\, du$$

Which is integrable by parts:

$$\langle v \rangle = \frac{a}{2b^2} = \frac{4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2}}{2\left(\frac{m}{k_B T}\right)^2}$$

$$= \sqrt{\frac{8k_B T}{\pi m}} = \frac{2}{\sqrt{\pi}} v_{\text{most probable}}$$

This gives an average speed which is just a little higher compared to the most probable speed.

4 Average Kinetic Energy

Since:

$$\langle E_k \rangle = \frac{1}{2}mv^2 \implies \langle E_k \rangle = \frac{1}{2}m\langle v^2 \rangle$$

We need $\langle v^2 \rangle$:

$$\langle v^2 \rangle = \int_0^\infty v^2 \Pr(v)\, dv = a\int_0^\infty v^4 e^{-bv^2/2}\, dv$$

Where we reuse $a = 4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2}$ and let $\alpha = \frac{m}{2k_B T}$, so the exponent becomes $-\alpha v^2$:

$$\langle v^2 \rangle = a\int_0^\infty v^4 e^{-\alpha v^2}\, dv$$

We apply Feynman's trick with $n = 2$:

$$\int_0^\infty e^{-\alpha x^2} x^{2n}\, dx = \frac{1}{2}\left(-\frac{\partial}{\partial \alpha}\right)^n \sqrt{\frac{\pi}{\alpha}}$$

$$-\frac{\partial}{\partial \alpha}\sqrt{\frac{\pi}{\alpha}} = \frac{1}{2}\sqrt{\pi}\,\alpha^{-3/2}$$

$$\left(-\frac{\partial}{\partial \alpha}\right)^2 \sqrt{\frac{\pi}{\alpha}} = -\frac{\partial}{\partial \alpha}\left[\frac{1}{2}\sqrt{\pi}\,\alpha^{-3/2}\right] = \frac{3}{4}\sqrt{\pi}\,\alpha^{-5/2}$$

Hence:

$$\int_0^\infty v^4 e^{-\alpha v^2}\, dv = \frac{3}{8}\sqrt{\pi}\,\alpha^{-5/2}$$

Substituting back with $\alpha = \frac{m}{2k_B T}$:

$$\langle v^2 \rangle = 4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2} \cdot \frac{3}{8}\sqrt{\pi}\left(\frac{m}{2k_B T}\right)^{-5/2}$$

$$= \frac{3\pi^{3/2}}{2} \cdot \frac{m^{3/2}}{(2\pi)^{3/2}(k_B T)^{3/2}} \cdot \frac{(2k_B T)^{5/2}}{m^{5/2}} = \frac{3}{2} \cdot 2 \cdot \frac{k_B T}{m} = \frac{3k_B T}{m}$$

Therefore:

$$\langle E_k \rangle = \frac{1}{2}m\langle v^2 \rangle = \frac{1}{2}m \cdot \frac{3k_B T}{m} = \boxed{\frac{3}{2}k_B T}$$

This lets us define the RMS (root mean square) speed:

$$v_{\text{rms}} = \sqrt{\langle v^2 \rangle} = \sqrt{\frac{3k_B T}{m}} = \sqrt{\frac{3}{2}}\, v_{\text{most probable}}$$

Summarising the three characteristic speeds of the distribution:

$$v_{\text{most probable}} = \sqrt{\frac{2k_BT}{m}} < \langle v \rangle = \sqrt{\frac{8k_BT}{\pi m}} < v_{\text{rms}} = \sqrt{\frac{3k_BT}{m}}$$

Tue 24 Mar 2026 12:00

Lecture 19 - Statistical Physics VI: Energies

1 Average Kinetic Energy

In 3D, we derived last lecture:

$$\langle E_k \rangle = \frac{3}{2} k_B T$$

Each component contributes equally, so:

$$E_{kx} = E_{ky} = E_{kz} = \frac{1}{2} k_B T$$

Remember from the second module of the course we have:

$$c = \frac{dQ}{dT}$$

And the first law of thermodynamics:

$$dU = dQ - pdV \implies dQ = dU + pdV$$

At a constant pressure or volume:

$$C_v = \left(\frac{\partial Q}{\partial T}\right)_V = \left(\frac{\partial U}{\partial T}\right)_V$$

We can define an additional value called Enthalpy (free energy):

$$dH = dU + PdV$$

Hence:

$$C_p = \left(\frac{\partial Q}{\partial T}\right)_P = \left(\frac{\partial H}{\partial T}\right)_V$$

For ideal gases, internal energy comes solely from kinetic energy. As we now have average kinetic energy as a function of T, we can now say (for per-atom and per-mole cases respectively):

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{3}{2} k_B$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{3}{2} R$$

This works perfectly for monatomic gases, where Mayer's relation also lets us define C_P:

Gas	C_V (J mol^{-1} K^{-1})	C_P (J mol^{-1} K^{-1})	$C_P - C_V$ (J mol^{-1} K^{-1})
Helium (He)	12.52	20.79	8.27
Argon (Ar)	12.45	20.79	8.34
Krypton (Kr)	12.45	20.79	8.34
Xenon (Xe)	12.52	20.79	8.27
Hydrogen (H_2)	20.42	28.74	8.32
Nitrogen (N_2)	20.76	29.07	8.31
Oxygen (O_2)	20.85	29.16	8.31
Carbon monoxide (CO)	20.85	29.16	8.31

Figure 19.1

But evidently breaks for diatomic molecules... They seem to have $C_V = 5R/2$? We'll investigate why later.

2 Law of Equipartition of Energy

The Law of Equipartition says that any substance in equilibrium at temperature T has an average energy of $k_BT/2$ per **degree of freedom**.

A **degree of freedom** is any term in the expression for the energy of the system which has a squared position or squared velocity term.

In 1D, our expression for kinetic energy is:

$$E_k = \frac{1}{2}mv_x^2 = \frac{1}{2}k_BT \implies C_V = \frac{1}{2}R$$

And in 3D:

$$E_k = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 = \frac{3}{2}k_BT \implies C_V = \frac{3}{2}R$$

2.1 Example

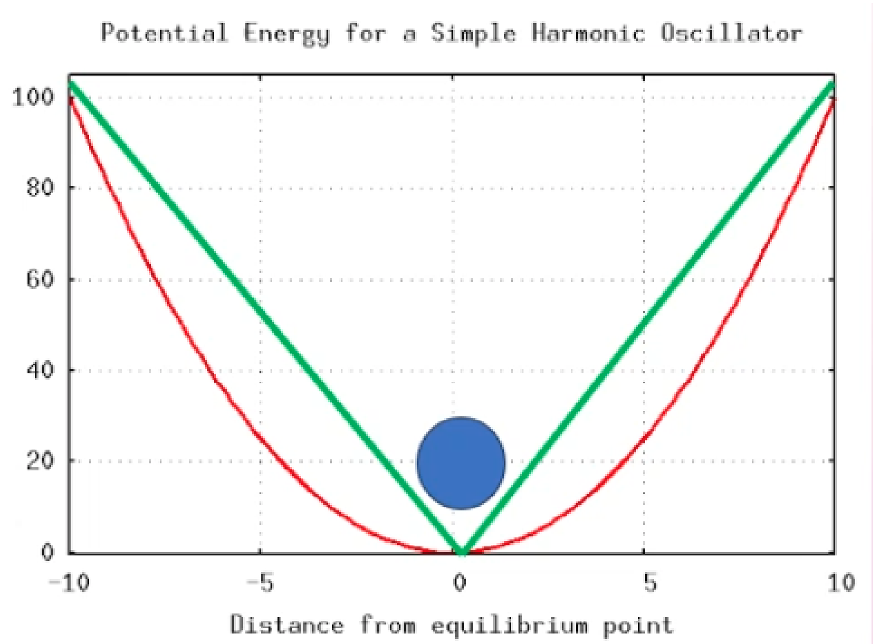

Figure 19.2

Consider a gas in a harmonic potential, acting as an oscillator in 1D:

$$E(V_x, x) = \frac{1}{2}mv_x^2 + \frac{1}{2}kx^2 = k_BT$$

Or in a linear potential:

$$E(v_x, x) = \frac{1}{2}mv_x^2 + Kx = \frac{1}{2}k_BTs$$

2.2 Dulong-Petit Heat Capacity of a Solid

We model the inter-atomic forces between atoms as springs with spring constant K. Each spring has vibrational energy given by:

$$E = \frac{1}{2}kx^2 = \frac{1}{2}mv^2 = k_BT$$

If we assume that the substance forms a perfect cubic lattice where every atom is connected to six other atoms, each spring is shared by two atoms so we say:

$$E = 3k_BT \implies c_V = 3R \approx 24.9\text{mol}^{-1}\text{K}^{-1}$$

This ends up being a surprisingly good fit for pure elements:

Element	Molar heat capacity C_V (J mol^{-1} K^{-1})
Bismuth	25.64
Gold	24.79
Platinum	25.04
Tin	25.30
Zinc	25.01
Gallium	24.60
Copper	25.14
Nickel	25.56
Iron	24.98
Calcium	24.67
Sulphur	25.30

Figure 19.3

3 Diatomic Molecules

Diatomic molecules can be modelled as being connected by a single spring:

$$E_{\text{vibrational}} = \frac{1}{2}kx^2 + \frac{1}{2}mv_x^2 \qquad \text{(2 DoF)}$$

However, they are also free to move around the volume in which they occupy, this gives some extra translational energy:

$$E_{\text{KE}} = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 \qquad \text{(3 DoF)}$$

Finally, they're also able to rotate around the axis joining them (perpendicularly):

$$E_{\text{rot}} = \frac{1}{2}I\omega_y^2 + \frac{1}{2}I\omega_z^2 \qquad \text{(2 DoF)}$$

In theory this means that a diatomic molecule has 7 degrees of freedom. We saw earlier however that the values for C_V for a diatomic molecule is $C_V = 5R/2$, only implying 5 DoF? Lets look at some actual data:

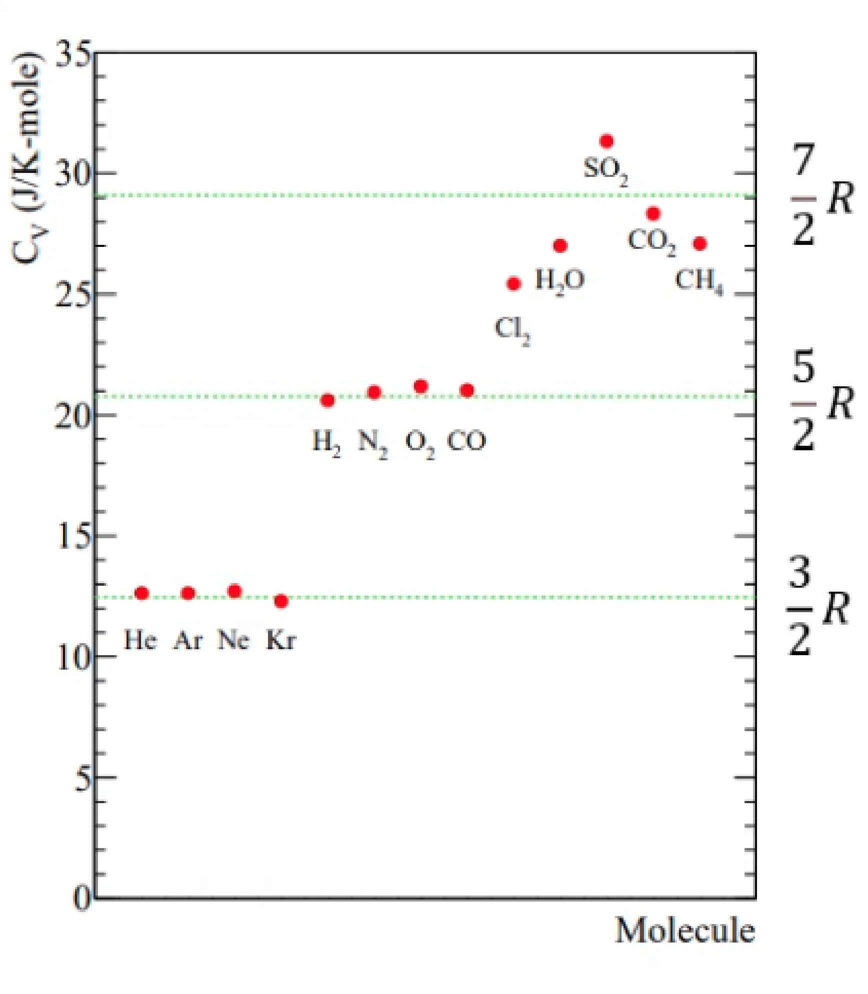

Figure 19.4

Our monatomic gases behave well, with values close to $3R/2$, but our diatomic gases seem to be more varied, with Chlorine having a much higher value between $5R/2$ and very roughly $7R/2$?

We can't say that our assumption of 7DoF is wrong, because Chlorine sits nicely between them. It means that this cannot, for example, simply lack rotational energy.

The states where a molecule has vibrational energy occurs at larger energies. In order to get a reasonable probability of measuring a state with a certain energy from the Boltzmann factor:

$$\Pr(E_i) \propto \exp\left(-\frac{E_i}{k_B T}\right)$$

We require $k_B T \lll E_i$. This gives a transition, where for some temperatures there is some intermediary probability where there's a good chance of measuring particles with and without sufficient energy for the final degrees of freedom.

This gives a plot like this:

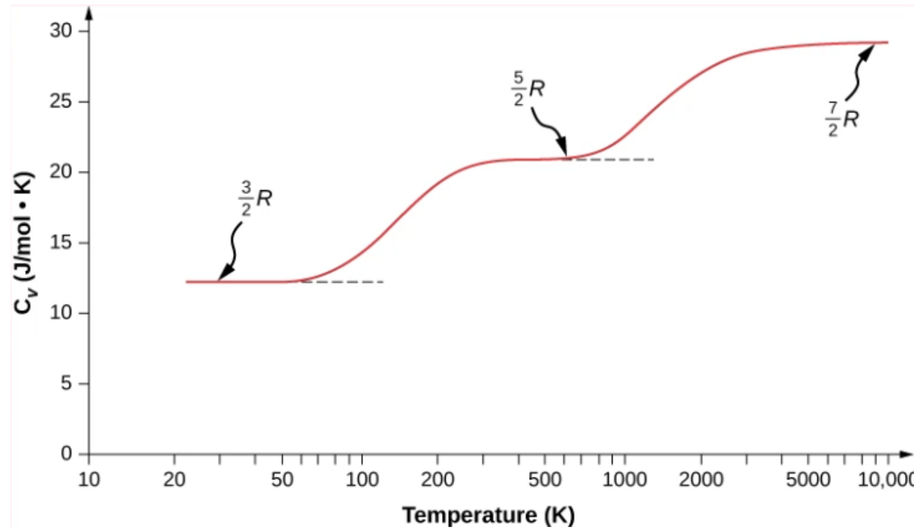

Figure 19.5

The translational component is always present, as our gas molecules are always present. The vibrational and rotational modes only appear at certain levels where there is sufficient energy to "unlock" them.

The degrees of freedom can be considered fully excited if:

$$k_B T \gg E_1 - E_0$$

And if:

$$k_B T \ll E_1 - E_0$$

There is insufficient energy to expect to encounter any molecules with sufficient energy for the motion associated with the degree of freedom, and it is said to be frozen out.

Thu 26 Mar 2026 13:00

Lecture 20 - Fluid Dynamics

1 Fluid Dynamics

Say we have no applied force, and the pressure at the liquid surface (for this weirdly shaped container) is P_0. What is the pressure P at some depth h?

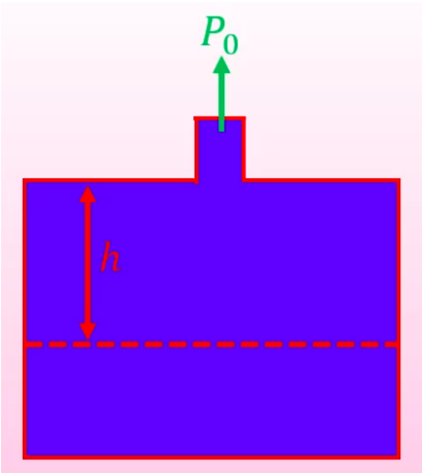

Figure 20.1

$$P = P_0 + \rho g h$$

If, and only if, the fluid has constant density ρ. If ρ varied as a function of height, we'd need to start integrating.

What if we applied a force to the liquid at the top of the spout?

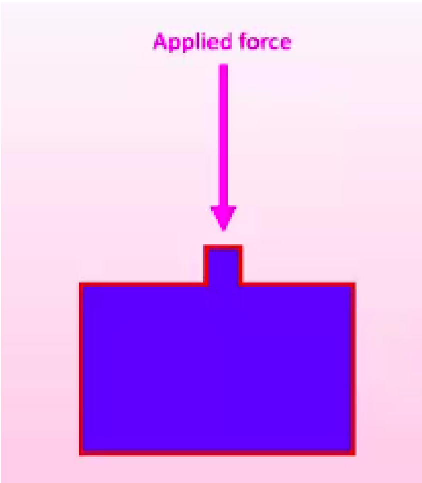

Figure 20.2

Pressure in the liquid will increase, so the applied force to the walls of the container will increase. Pascal's law says that this increase in pressure is applied to every part of the fluid (and the walls of the enclosure) equally and undiminished.

1.1 Example: Hydraulic Presses

If we apply a force F_1 to a liquid in some very small cross-sectional area a_1, then the pressure this results in must be applied to all points in the liquid:

$$P = \frac{F_1}{a_1}$$

Therefore, on the "output" of the press:

$$P = \frac{F_2}{a_2}$$

If a_2 is much larger than a_1, the output force F_2 must be proportionally larger than the input force F_1 such that:

$$F_2 = \frac{F_1 a_2}{a_1}$$

This effectively lets us magnify forces.

1.2 Fluid in a Pipe

In a steady state (a state with does not change with respect to time) the amount of water that passes a given point must also be constant with respect to time.

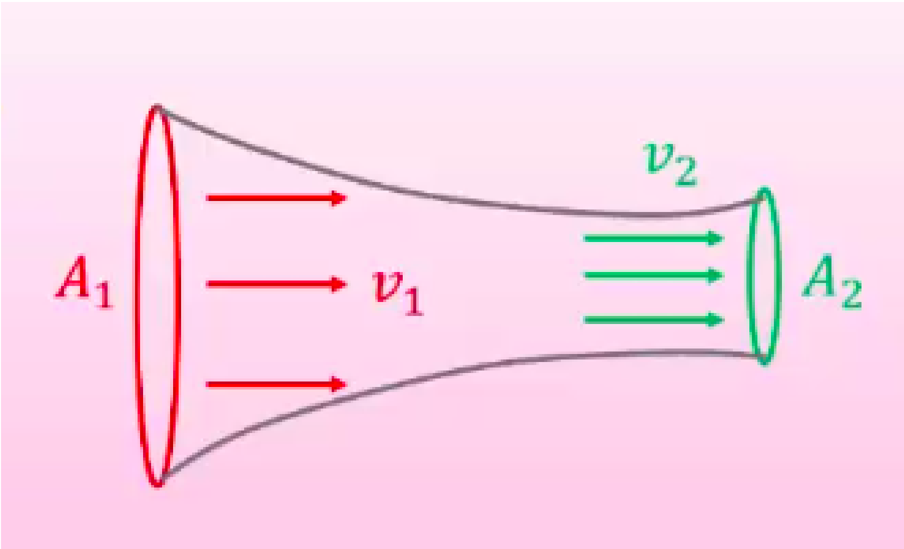

Figure 20.3

For this to be a steady state, the amount of liquid passing both points must be the same (otherwise liquid would get stuck and build up). Therefore, we must see:

$$A_1 v_1 = A_2 v_2$$

For $v_1 \neq v_2$, a force must be applied to the liquid to change velocity. This force comes from the neighbouring fluid molecules.

Previously, we saw:

$$P = P_0 + \rho g h$$

To account for the liquid now being non-stationary and having a velocity, we need to add a term accounting for pressure from kinetic energy:

$$P + \rho g h + \frac{1}{2}\rho v^2$$

Here, P is called static pressure, $\rho g h$ is called hydrostatic pressure and $\frac{1}{2}\rho v^2$ is called dynamic pressure. This full equation becomes useful in a context like this:

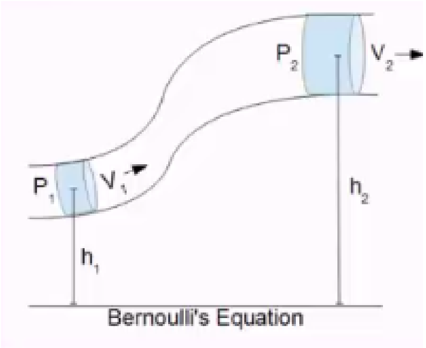

Figure 20.4

End of Module